PhenObs Cluster 1 analyses

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Load libraries needed

```
library(readr)
library(here)
```

```
## here() starts at C:/Users/jimenezalfaro/OneDrive - Universidad de Oviedo/IMIB/Analyses/PhenObs_Cluste
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v purrr     1.2.2
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(purrr)
library(sf)
```

```
## Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

```r
library(glmmTMB)
```

```
## Warning in check_dep_version(dep_pkg = "TMB"): package version mismatch:
## glmmTMB was built with TMB package version 1.9.21
## Current TMB package version is 1.9.23
## Please re-install glmmTMB from source or restore original 'TMB' package (see '?reinstalling' for more
```

```r
library(ggeffects)
library(taxize)
library(rgbif)
library(readxl)
library(gridExtra)
```

```
##
## Adjuntando el paquete: 'gridExtra'
##
## The following object is masked from 'package:dplyr':
##
##     combine
```

```r
library(furrr)
```

```
## Cargando paquete requerido: future
##
## Adjuntando el paquete: 'future'
##
## The following object is masked from 'package:taxize':
##
##     resolve
```

```r
library(countrycode)
library(progress)
library(ggrepel)
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```r
library(vegan)
```

```
## Cargando paquete requerido: permute
```

```r
library(broom)
library(broom.mixed)
library(rnaturalearth)
library(units)
```

```
## udunits database from C:/Users/jimenezalfaro/AppData/Local/R/win-library/4.6/units/share/udunits/udu
```

```r
library(sysfonts)
library(showtext)
```

```
## Cargando paquete requerido: showtextdb
```

```r
library(reproduciblemanuscript)
library(modelsummary)
library(patchwork)
library(flextable)
```

```
##
## Adjuntando el paquete: 'flextable'
##
## The following object is masked from 'package:purrr':
##
##     compose
```

```r
library(officer)
```

```
##
## Adjuntando el paquete: 'officer'
##
## The following object is masked from 'package:readxl':
##
##     read_xlsx
```

Set ggplot theme

```r
font_add("Times", "C:/Windows/Fonts/times.ttf")
showtext_auto()
```

```r
windowsFonts(Times = windowsFont("Times New Roman"))
```

```r
theme_set(
  theme_bw(base_family = "Times") +
    theme(
      text = element_text(size = 30),
      axis.title = element_text(size = 30),
      axis.text = element_text(size = 24),
      plot.title = element_text(size = 30)
    )
)
```

Load previously created objects

```r
load(here("objects", "sens_per_endanger_RL_Staude_Europe.Rdata"))
load(here("objects", "sens_per_endanger_stress_RL_Staude_Europe.Rdata"))
load(here("objects", "sens_per_endanger_stress_ell_RL_Staude_Europe.Rdata"))
```

Read data

Red List

Load Red List data (MOTIVATE subset).

```r
redlist <- read_csv(here("data", "raw", "Staude_et_al_2025", "redlist_checked.csv"))
```

```
## Rows: 63958 Columns: 9
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (8): accepted_taxon_name, accepted_taxon_authors, status, Redlist_cat, I...
## dbl (1): Publ_year
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
redlist
```

```
## # A tibble: 63,958 x 9
##    accepted_taxon_name       accepted_taxon_authors status Redlist_cat ISO3_Code
##    <chr>                     <chr>                  <chr>  <chr>       <chr>
##  1 Abies × borisii-regis     Mattf.                 threa~ VU          ALB
##  2 Acantholimon albanicum    O.Schwarz & F.Mey.     threa~ EN          ALB
##  3 Acer hyrcanum             Fisch. & C.A.Mey.      threa~ EN          ALB
```

```
##  4 Achillea clusiana         Tausch                  threa~ CR          ALB
##  5 Achillea grandifolia      Friv.                   threa~ CR          ALB
##  6 Acis ionica               Bareka, Kamari & Phit~ threa~ CR          ALB
##  7 Aconitum lamarckii        Rchb. ex Spreng.        threa~ VU          ALB
##  8 Adiantum capillus-veneris L.                      threa~ VU          ALB
##  9 Aesculus hippocastanum    L.                      threa~ CR          ALB
## 10 Ajuga piskoi              Degen & Bald.           threa~ EN          ALB
## # i 63,948 more rows
## # i 4 more variables: ISO2_Code <chr>, Country <chr>, region_name <chr>,
## #   Publ_year <dbl>
```

Sensitivity

Read sensitivity data (slopes from Robert).

```
slopes_FlOn <- read_csv(here("data", "raw", "slopes_20260302", "Overview_Slopes_all_New_FlOn.csv")) %>%
  dplyr::select(-...1) %>% mutate(stage = "FlOn")
```

```
## New names:
## Rows: 171 Columns: 11
## -- Column specification
## -------------------------------------------------------- Delimiter: "," chr
## (1): Species dbl (10): ...1, mean_slope, ci_lower, ci_upper, slope_var,
## n_gardens, temp_m...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

```
slopes_SensOn <- read_csv(here("data", "raw", "slopes_20260302", "Overview_Slopes_all_SenOn.csv")) %>%
  dplyr::select(-...1) %>% mutate(stage = "SensOn")
```

```
## New names:
## Rows: 156 Columns: 11
## -- Column specification
## -------------------------------------------------------- Delimiter: "," chr
## (1): Species dbl (10): ...1, mean_slope, ci_lower, ci_upper, slope_var,
## n_gardens, temp_m...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

```
slopes_LeafUnf <- read_csv(here("data", "raw", "slopes_20260302", "Overview_Slopes_all_LeafUnf.csv")) %
  dplyr::select(-...1) %>% mutate(stage = "LeafUnf")
```

```
## New names:
## Rows: 175 Columns: 11
## -- Column specification
## -------------------------------------------------------- Delimiter: "," chr
## (1): Species dbl (10): ...1, mean_slope, ci_lower, ci_upper, slope_var,
## n_gardens, temp_m...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

```r
slopes_GSL <- read_csv(here("data", "raw", "slopes_20260302", "Overview_Slopes_all_GSL.csv")) %>%
  dplyr::select(-...1) %>% mutate(stage = "GSL")
```

```
## New names:
## Rows: 154 Columns: 11
## -- Column specification
## -------------------------------------------------------- Delimiter: "," chr
## (1): Species dbl (10): ...1, mean_slope, ci_lower, ci_upper, slope_var,
## n_gardens, temp_m...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

```r
slopes_FrOn <- read_csv(here("data", "raw", "slopes_20260302", "Overview_Slopes_all_New_FrOn.csv")) %>%
  dplyr::select(-...1) %>% mutate(stage = "FrOn")
```

```
## New names:
## Rows: 149 Columns: 11
## -- Column specification
## -------------------------------------------------------- Delimiter: "," chr
## (1): Species dbl (10): ...1, mean_slope, ci_lower, ci_upper, slope_var,
## n_gardens, temp_m...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

```r
slopes_FlDur <- read_csv(here("data", "raw", "slopes_20260302", "Overview_Slopes_all_New_FlDur.csv")) %
  dplyr::select(-...1) %>% mutate(stage = "FlDur")
```

```
## New names:
## Rows: 171 Columns: 11
## -- Column specification
## -------------------------------------------------------- Delimiter: "," chr
## (1): Species dbl (10): ...1, mean_slope, ci_lower, ci_upper, slope_var,
## n_gardens, temp_m...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

```r
slopes_FlPeak <- read_csv(here("data", "raw", "slopes_20260302", "Overview_Slopes_all_New_FlPeak.csv"))
  dplyr::select(-...1) %>% mutate(stage = "FlPeak")
```

```
## New names:
## Rows: 169 Columns: 11
## -- Column specification
## -------------------------------------------------------- Delimiter: "," chr
## (1): Species dbl (10): ...1, mean_slope, ci_lower, ci_upper, slope_var,
## n_gardens, temp_m...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

```
slopes_FlOn
```

```
## # A tibble: 171 x 11
##    Species    mean_slope ci_lower ci_upper slope_var n_gardens temp_min temp_max
##    <chr>           <dbl>    <dbl>    <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
##  1 Achillea ~     -3.48   -10.5     10.1      26.4          13     5.44     14.6
##  2 Aconitum ~      8.37     1.33    16.9      14.7           6     8.66     12.9
##  3 Acorus ca~     -8.58   -22.9      1.76     41.8           6     9.18     12.0
##  4 Actaea sp~      3.94     0.322    6.65      2.69          4     8.56     12.6
##  5 Adonis ve~     -1.82    -8.15     6.30     13.3           6     8.11     11.1
##  6 Aegopodiu~     -2.69    -6.58     0.416     2.70          7     4.29     12.0
##  7 Ajuga pyr~     -0.622   -4.49    10.7      26.8           3     3.87     10.9
##  8 Alchemill~      6.80    -0.342   18.2      23.5           3     8.73     11.5
##  9 Alchemill~     -6.15   -13.0      1.13     12.8           4     7.76     10.9
## 10 Allium sc~     -5.54    -6.79    -3.78      0.526         5     3.49     14.6
## # i 161 more rows
## # i 3 more variables: temp_mean <dbl>, n_data <dbl>, stage <chr>
```

```
slopes_SensOn
```

```
## # A tibble: 156 x 11
##    Species    mean_slope ci_lower ci_upper slope_var n_gardens temp_min temp_max
##    <chr>           <dbl>    <dbl>    <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
##  1 Achillea ~    -13.9     -23.9     -5.39      27.2        11     8.64     16.2
##  2 Aconitum ~      5.09     -6.79    16.7       28.9         5     8.64     12.4
##  3 Acorus ca~     -7.20    -23.7      5.42      51.7         7     8.64     12.2
##  4 Actaea sp~      9.27     -4.06    20.0       40.8         4     9.85     12.8
##  5 Adonis ve~     -0.935   -16.7     11.0       49.4         6     8.64     12.8
##  6 Aegopodiu~      7.07     -7.94    22.0       51.7         8     4.69     12.2
##  7 Ajuga rep~      4.20     -7.58    12.5       32.0         3    10.4      16.2
##  8 Alchemill~      6.82    -16.6     34.6      176.          4     8.64     11.6
##  9 Allium sc~     -6.24    -19.7     37.4      171.          3     9.58     15.5
## 10 Allium ur~     -4.86    -11.3      2.70      12.5        10     9.58     15.4
## # i 146 more rows
## # i 3 more variables: temp_mean <dbl>, n_data <dbl>, stage <chr>
```

```
slopes_LeafUnf
```

```
## # A tibble: 175 x 11
##    Species    mean_slope ci_lower ci_upper slope_var n_gardens temp_min temp_max
##    <chr>           <dbl>    <dbl>    <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
##  1 Achillea ~      -2.27   -13.2     8.48      33.1         13     4.86     16.2
##  2 Aconitum ~      -8.12   -13.5    -1.05      10.4          5     8.64     12.8
##  3 Acorus ca~      -7.58   -23.9     0.867     52.0          8     4.83     12.2
##  4 Actaea sp~      13.3      4.78   24.3       26.1          4     8.64     12.4
##  5 Adonis ve~      -6.78   -13.1    -0.0753     9.90         6     8.64     12.8
##  6 Aegopodiu~      -7.69   -20.3     2.98      32.6          7     8.64     12.2
##  7 Ajuga pyr~      -6.38   -12.5    38.5      173.           3     4.69     11.9
##  8 Alchemill~     -15.0    -29.6    -4.52      38.7          4     8.64     11.9
##  9 Allium sc~      -9.17   -12.0    -6.02       2.51         4     3.58     15.5
## 10 Allium ur~      -6.49   -10.7     0.779      7.44        11     4.83     15.5
## # i 165 more rows
## # i 3 more variables: temp_mean <dbl>, n_data <dbl>, stage <chr>
```

```
slopes_GSL
```

```
## # A tibble: 154 x 11
##    Species    mean_slope ci_lower ci_upper slope_var n_gardens temp_min temp_max
##    <chr>           <dbl>    <dbl>    <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
##  1 Achillea ~     -12.2   -22.2       5.00      44.8        11     8.64     15.8
##  2 Aconitum ~      11.7     1.35     20.1       23.0         5     8.64     12.4
##  3 Acorus ca~       3.75  -17.4      20.7       85.6         8     4.83     12.2
##  4 Actaea sp~      -7.48  -24.1      10.9       85.3         4     8.64     12.4
##  5 Adonis ve~       4.57  -10.8      15.3       43.6         6     8.64     12.8
##  6 Aegopodiu~      15.5    -4.36     35.0       84.7         8     4.86     12.2
##  7 Alchemill~      21.7     0.516    48.6      149.          4     8.64     11.6
##  8 Allium sc~      -2.14  -19.3      22.3      169.          3     9.58     15.5
##  9 Allium ur~       1.92   -6.74     12.0       21.8        10     8.64     15.5
## 10 Allium vi~       7.15   -8.55     14.3       28.8         7     4.83     12.2
## # i 144 more rows
## # i 3 more variables: temp_mean <dbl>, n_data <dbl>, stage <chr>
```

```
slopes_FrOn
```

```
## # A tibble: 149 x 11
##    Species    mean_slope ci_lower ci_upper slope_var n_gardens temp_min temp_max
##    <chr>           <dbl>    <dbl>    <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
##  1 Achillea ~     -9.82   -31.2      0.898     58.6         11     10.1      17.4
##  2 Aconitum ~      6.12    -0.795   15.0       17.1          6     10.1      13.9
##  3 Acorus ca~     12.4     -4.53    27.2       52.5          7     10.1      13.5
##  4 Actaea sp~      1.70    -4.04     7.51       9.89         4      9.25     13.6
##  5 Adonis ve~     -1.26    -6.76     3.94       7.87         6      8.93     12.2
##  6 Aegopodiu~     -3.48   -13.6      3.29      18.2          6     10.4      13.1
##  7 Alchemill~      2.08    -4.29     9.52      13.0          3     11.0      12.9
##  8 Alchemill~      0.905   -5.78     5.96       9.14         4     10.2      12.2
##  9 Allium sc~     -2.51    -6.03     2.24       4.15         5      4.95     15.0
## 10 Allium ur~     -5.54    -9.67     1.98       9.94        10      8.93     14.8
## # i 139 more rows
## # i 3 more variables: temp_mean <dbl>, n_data <dbl>, stage <chr>
```

```
slopes_FlDur
```

```
## # A tibble: 171 x 11
##    Species    mean_slope ci_lower ci_upper slope_var n_gardens temp_min temp_max
##    <chr>           <dbl>    <dbl>    <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
##  1 Achillea ~      11.6     -2.62   24.8       46.4         11     7.9      16.7
##  2 Aconitum ~      -3.34   -13.0     3.22      15.2          6     7.28     10.9
##  3 Acorus ca~       5.55    -4.58   14.8       24.5          6     7.89     10.8
##  4 Actaea sp~     -13.2    -18.9    -5.26      10.8          4     8.22     11.9
##  5 Adonis ve~      -6.83   -13.4    -1.31       9.45         6     8.23     11.1
##  6 Aegopodiu~      -2.37    -6.28   -0.593      1.95         7     3.65     10.8
##  7 Agrimonia~      -4.52    -8.18   -1.19       2.98         3     6.83     13.6
##  8 Ajuga pyr~       1.47    -2.35    6.75       7.98         3     4        11.4
##  9 Alchemill~      58.0     43.8    77.5       67.9          4     8.62     12.6
## 10 Allium sc~      15.4     11.3    19.2        4.16         5     2.53     16.5
## # i 161 more rows
## # i 3 more variables: temp_mean <dbl>, n_data <dbl>, stage <chr>
```

```
slopes_FlPeak
```

```
## # A tibble: 169 x 11
##    Species    mean_slope ci_lower ci_upper slope_var n_gardens temp_min temp_max
##    <chr>           <dbl>    <dbl>    <dbl>     <dbl>     <dbl>    <dbl>    <dbl>
##  1 Achillea ~      -3.02   -9.68    8.31      20.1          12     9.74     16.2
##  2 Aconitum ~       7.46    2.72   12.6        7.30          6     9.18     13.6
##  3 Acorus ca~      -8.65  -17.6    -0.551     18.4           6     8.25     11.6
##  4 Actaea sp~       1.02   -4.56    4.21       4.74          4     7.57     12.6
##  5 Adonis ve~      -3.66  -10.3     4.63      14.2           6     8        11.0
##  6 Aegopodiu~      -3.21   -6.13    0.719      3.32          7     5.12     12.0
##  7 Alchemill~       5.93   -0.623  14.2       14.8           3     8.91     11.8
##  8 Alchemill~      -3.10   -8.00    3.90      10.4           4     8.25     10.9
##  9 Allium sc~      -3.87   -5.24   -2.57       0.456         5     4.05     14.9
## 10 Allium ur~      -5.95   -9.50    0.0617     5.69         11     5.31     14.7
## # i 159 more rows
## # i 3 more variables: temp_mean <dbl>, n_data <dbl>, stage <chr>
```

Join and get unique species

```
slopes_all <- bind_rows(slopes_FlOn, slopes_SensOn, slopes_LeafUnf, slopes_GSL,
                        slopes_FrOn, slopes_FlDur, slopes_FlPeak)
```

```
slopes_all_species <- slopes_all %>% distinct(Species)
```

Stress tolerance

Read stress tolerance scores.

```
stress <- read_delim(here("data", "raw", "Stress_Tolerance.csv"), delim = ";") %>%
  rename(stress_tol = `Stress tolerator score_percentage`)
```

```
## Rows: 263 Columns: 2
## -- Column specification --------------------------------------------------------
## Delimiter: ";"
## chr (1): Species
## dbl (1): Stress tolerator score_percentage
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
stress
```

```
## # A tibble: 263 x 2
##    Species               stress_tol
##    <chr>                      <dbl>
##  1 Achillea millefolium        44.4
##  2 Aconitum lycoctonum         13.4
##  3 Acorus calamus              23.8
```

```
##  4 Actaea spicata              11.1
##  5 Adonis amurensis            24.4
##  6 Adonis vernalis             41.5
##  7 Aegopodium podagraria        6.7
##  8 Agrimonia eupatoria         41.6
##  9 Agrimonia pilosa            53.7
## 10 Ajuga pyramidalis           34.1
## # i 253 more rows
```

Ellenberg tendency of change

Read Ellenberg tendency of change data.

```
ellenberg <- read_excel(here("data", "raw", "IndicatorValues_Ellenberg.xlsx"))
```

```
## Warning: Expecting numeric in Q1081 / R1081C17: got '?'
```

```
## Warning: Expecting numeric in Q1287 / R1287C17: got '?'
```

```
## Warning: Expecting numeric in Q2746 / R2746C17: got '?'
```

```
ellenberg
```

```
## # A tibble: 2,792 x 19
##    NR_E  Art    GATT_NAME OEK_L OEK_T OEK_K OEK_F OEK_R OEK_N OEK_S LF_LEB LF_B
##    <chr> <chr>  <chr>     <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>  <chr>
##  1 101   Abies~ Abies     (3)   5     4     x     x     x     0     P      I
##  2 401   Acer ~ Acer      (5)   6     4     5     7     6     0     P      S
##  3 403   Acer ~ Acer      (6)   8     4     3     8     4     0     P      S
##  4 404   Acer ~ Acer      (5)   6     6     6     7     7     0     P      S
##  5 405   Acer ~ Acer      (5)   8     4     4     8     6     0     P      S
##  6 408   Acer ~ Acer      (4)   6     4     x     x     x     0     P      S
##  7 409   Acer ~ Acer      (4)   x     4     6     x     7     0     P      S
##  8 501   Acera~ Aceras    7     7     2     4~    8     3     0     G      S
##  9 601   Achil~ Achillea  9     2     4     5     8     3     0     H      W
## 10 623   Achil~ Achillea  8     6     6     8~    7     6     0     H      S
## # i 2,782 more rows
## # i 7 more variables: SZ_VERH_GR <chr>, SZ_VERH_K <chr>, SZ_O <dbl>,
## #   SZ_V <dbl>, HAEUF_M <dbl>, HAEUF_D <chr>, HAEUF_AE_change <dbl>
```

Resolve species names

Resolve species in all datasets to GBIF accepted names using the package taxize:

Species in sensitivity data

Species in Red List data

Resolve species (accepted_taxon_name) to GBIF accepted names:

Species in stress tolerance scores

Species in Ellenberg tendency of change data

```
ellenberg <- ellenberg %>%
  mutate(Art = ifelse(Art == "Primula veris (officinalis)#f)",
                      "Primula veris (officinalis)", Art))
```

Create occurrence matrix

Occurrence matrices are created using the package rgbif.

Get countries form RL

Get vector of ISO3 codes for European countries.

Build occurrence matrix

Build the species × country occurrence matrix with rgbif.

Settings:

Helper functions:

Query GBIF once per species:

Parallel querying:

Convert ISO2 to ISO3. GBIF returns country codes as ISO2.Keep Kosovo, discard unknown countries. Keep only countries in Europe.

Build complete species × country matrix:

Presence/Absence matrix:

Matching Phenobs - MOTIVATE Red List data

Use the RL from all available countries!

Try to match the full PhenObs spp list with the RL to see how many species are there in each category for each country.

Join RL and spp list in sensitivity data from PhenObs

Data prep RL

Verify that there is only one record per species and country:

Set those where status is not_evaluated_dd as not_threatened:

Join occurrence data and Red List data

Change occurrence from 1 to 0 when threat == "extinct" (so we believe that RL data is correct, although checking in GBIF, records of these spp in these countries are found recently in some cases).

Calculate measures of endangerment

Proportion of countries where the species is endangered, and proportion of range where species is endangered. Also, country_threat_index and range_threat_index (include severity of threat).

Add info on country areas:

Define threat scores:

Data prep sensitivity data

Prepare datasets for models

Sensitivity + % endangerment

Sensitivity + % endangerment + stress

Sensitivity + % endangerment + Ellenberg tendency of change

Deal with non-numeric values in indicator values

Columns OEK_L - OEK_S and HAEUF_AE_change

```
ellenberg <- ellenberg %>%
  mutate(across(OEK_L:OEK_S, ~ parse_number(.)))
```

```
## Warning: There were 7 warnings in `mutate()`.
## The first warning was:
## i In argument: `across(OEK_L:OEK_S, ~parse_number(.))`.
## Caused by warning:
## ! 28 parsing failures.
## row col expected actual
## 184  -- a number      x
## 200  -- a number      x
## 676  -- a number      x
## 744  -- a number      x
## 798  -- a number      x
## ... ... ........ ......
## See problems(...) for more details.
## i Run `dplyr::last_dplyr_warnings()` to see the 6 remaining warnings.
```

Join with sens_per_endanger:

Save datasets for later use

Reshape data

```
sens_endanger <- sens_per_endanger %>%
  pivot_longer(cols = matches("^(mean_slope|slope_var|sens_01|sens_sign)_"),
               names_to = c(".value", "stage"),
               names_pattern = "(mean_slope|slope_var|sens_01|sens_sign)_(.*)")
sens_endanger
```

```
## # A tibble: 1,302 x 15
##    species     total_countries threatened_countries perc_endangerment total_area
##    <chr>                 <int>                <int>              <dbl>      <dbl>
##  1 Achillea m~              39                    0                  0     22927719.
##  2 Achillea m~              39                    0                  0     22927719.
##  3 Achillea m~              39                    0                  0     22927719.
##  4 Achillea m~              39                    0                  0     22927719.
##  5 Achillea m~              39                    0                  0     22927719.
##  6 Achillea m~              39                    0                  0     22927719.
##  7 Achillea m~              39                    0                  0     22927719.
##  8 Aconitum l~              36                    3               8.33     22553518.
##  9 Aconitum l~              36                    3               8.33     22553518.
## 10 Aconitum l~              36                    3               8.33     22553518.
## # i 1,292 more rows
## # i 10 more variables: threatened_area <dbl>, perc_endangered_range <dbl>,
## #   country_threat_index <dbl>, range_threat_index <dbl>,
## #   prop_endangerment <dbl>, prop_endangered_range <dbl>, stage <chr>,
## #   mean_slope <dbl>, slope_var <dbl>, sens_01 <fct>
```

```
sens_endanger_stress <- sens_per_endanger_stress %>%
  pivot_longer(cols = matches("^(mean_slope|slope_var|sens_01|sens_sign)_"),
               names_to = c(".value", "stage"),
               names_pattern = "(mean_slope|slope_var|sens_01|sens_sign)_(.*)")
sens_endanger_stress
```

```
## # A tibble: 1,302 x 16
##    species     total_countries threatened_countries perc_endangerment total_area
##    <chr>                 <int>                <int>              <dbl>      <dbl>
##  1 Achillea m~              39                    0                  0     22927719.
##  2 Achillea m~              39                    0                  0     22927719.
##  3 Achillea m~              39                    0                  0     22927719.
##  4 Achillea m~              39                    0                  0     22927719.
##  5 Achillea m~              39                    0                  0     22927719.
##  6 Achillea m~              39                    0                  0     22927719.
##  7 Achillea m~              39                    0                  0     22927719.
##  8 Aconitum l~              36                    3               8.33     22553518.
##  9 Aconitum l~              36                    3               8.33     22553518.
## 10 Aconitum l~              36                    3               8.33     22553518.
## # i 1,292 more rows
## # i 11 more variables: threatened_area <dbl>, perc_endangered_range <dbl>,
## #   country_threat_index <dbl>, range_threat_index <dbl>,
```

```
## #   prop_endangerment <dbl>, prop_endangered_range <dbl>, stress_tol <dbl>,
## #   stage <chr>, mean_slope <dbl>, slope_var <dbl>, sens_01 <fct>
```

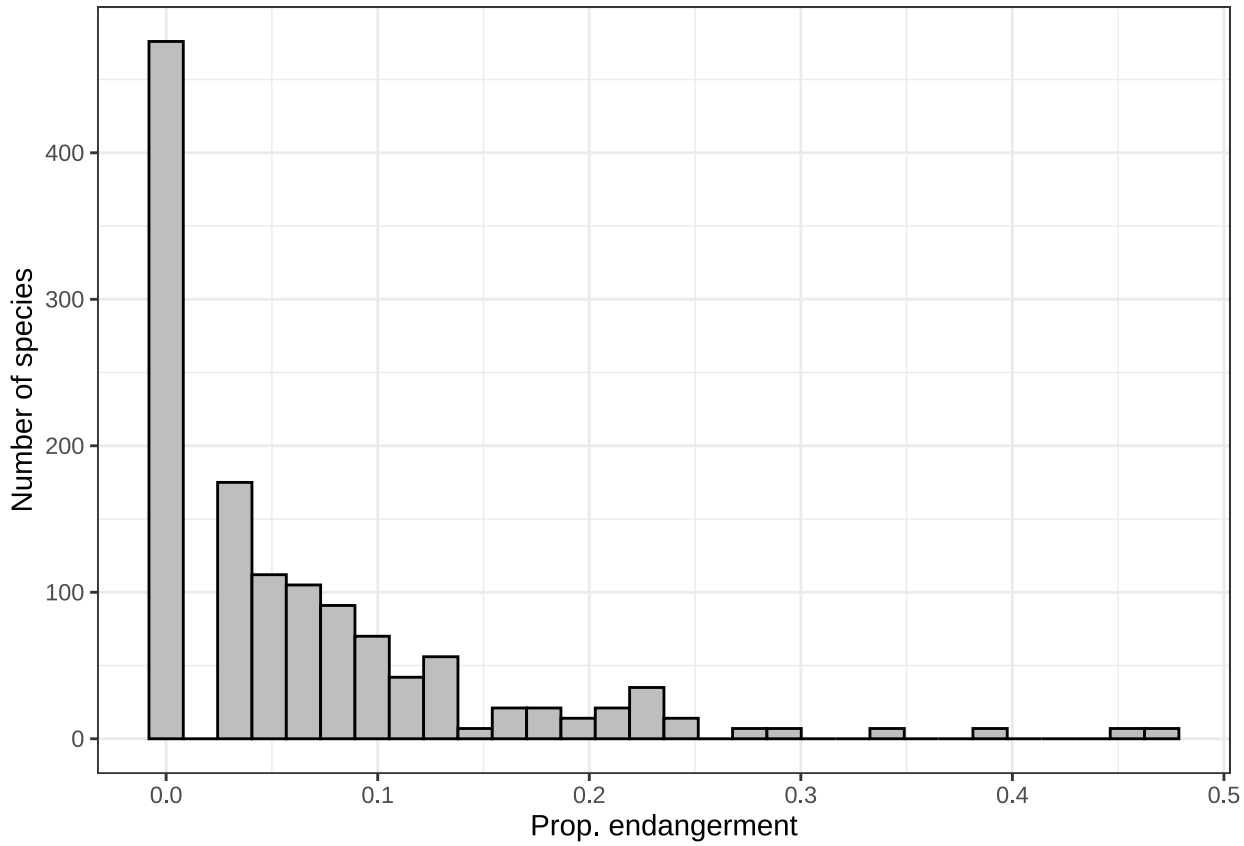
```
sens_endanger_stress_ell <- sens_per_endanger_stress_ell %>%
  pivot_longer(cols = matches("^(mean_slope|slope_var|sens_01|sens_sign)_"),
               names_to = c(".value", "stage"),
               names_pattern = "(mean_slope|slope_var|sens_01|sens_sign)_(.*)")
sens_endanger_stress_ell
```

```
## # A tibble: 1,302 x 17
##    species     total_countries threatened_countries perc_endangerment total_area
##    <chr>                 <int>                <int>              <dbl>      <dbl>
##  1 Achillea m~              39                    0                  0     22927719.
##  2 Achillea m~              39                    0                  0     22927719.
##  3 Achillea m~              39                    0                  0     22927719.
##  4 Achillea m~              39                    0                  0     22927719.
##  5 Achillea m~              39                    0                  0     22927719.
##  6 Achillea m~              39                    0                  0     22927719.
##  7 Achillea m~              39                    0                  0     22927719.
##  8 Aconitum l~              36                    3               8.33     22553518.
##  9 Aconitum l~              36                    3               8.33     22553518.
## 10 Aconitum l~              36                    3               8.33     22553518.
## # i 1,292 more rows
## # i 12 more variables: threatened_area <dbl>, perc_endangered_range <dbl>,
## #   country_threat_index <dbl>, range_threat_index <dbl>,
## #   prop_endangerment <dbl>, prop_endangered_range <dbl>, stress_tol <dbl>,
## #   change <int>, stage <chr>, mean_slope <dbl>, slope_var <dbl>, sens_01 <fct>
```

Distributions

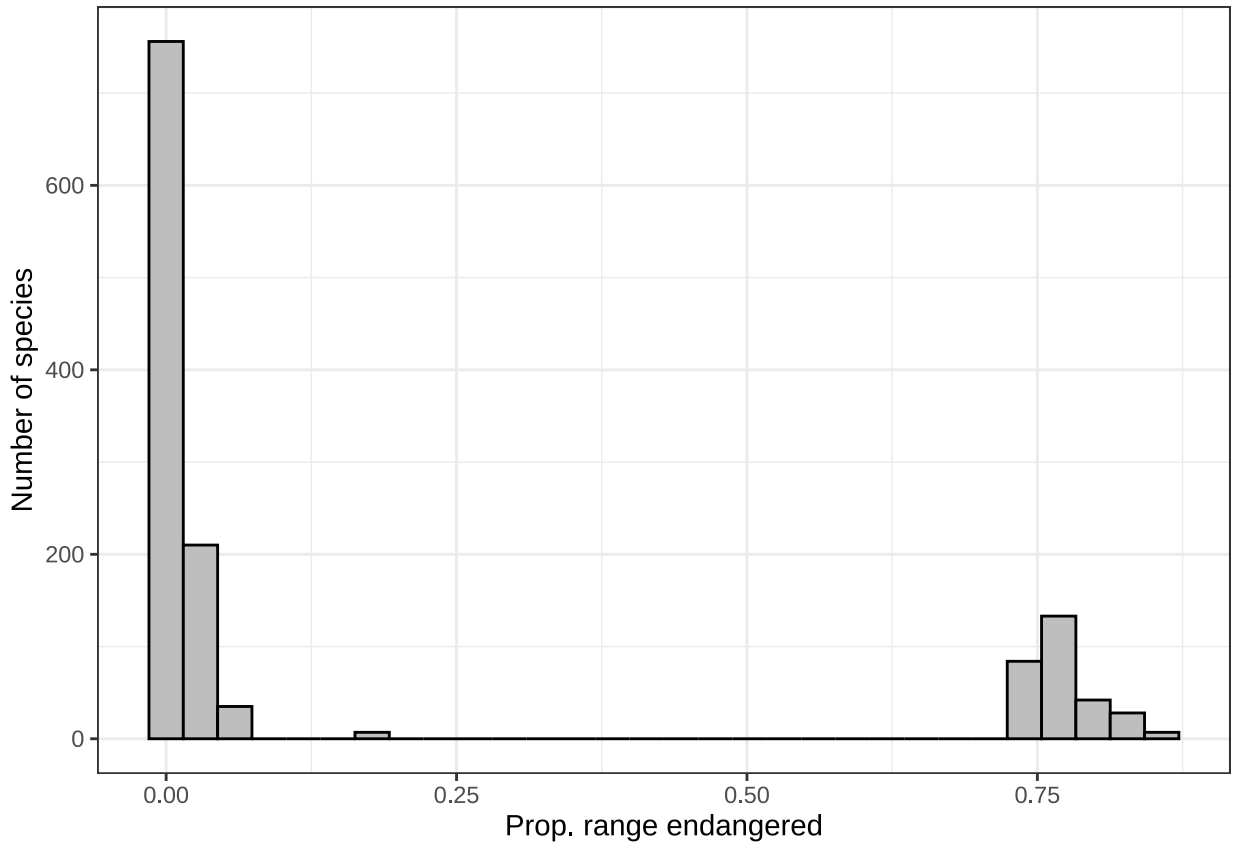
```
ggplot(sens_endanger, aes(x = prop_endangerment)) +
  geom_histogram(color = "black", fill = "grey") +
  labs(x = "Prop. endangerment", y = "Number of species") +
  theme_bw()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

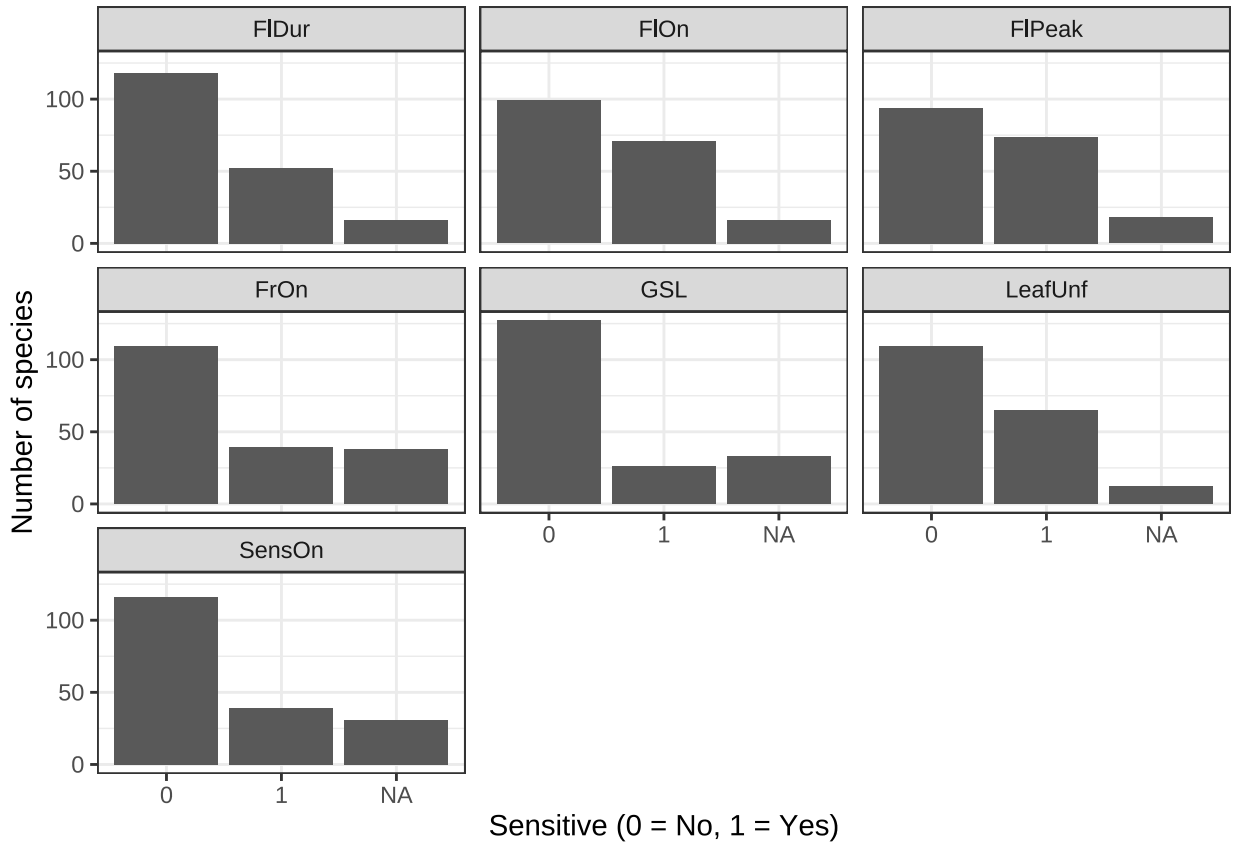



```
ggplot(sens_endanger, aes(x = prop_endangered_range)) +
  geom_histogram(color = "black", fill = "grey") +
  labs(x = "Prop. range endangered", y = "Number of species") +
  theme_bw()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



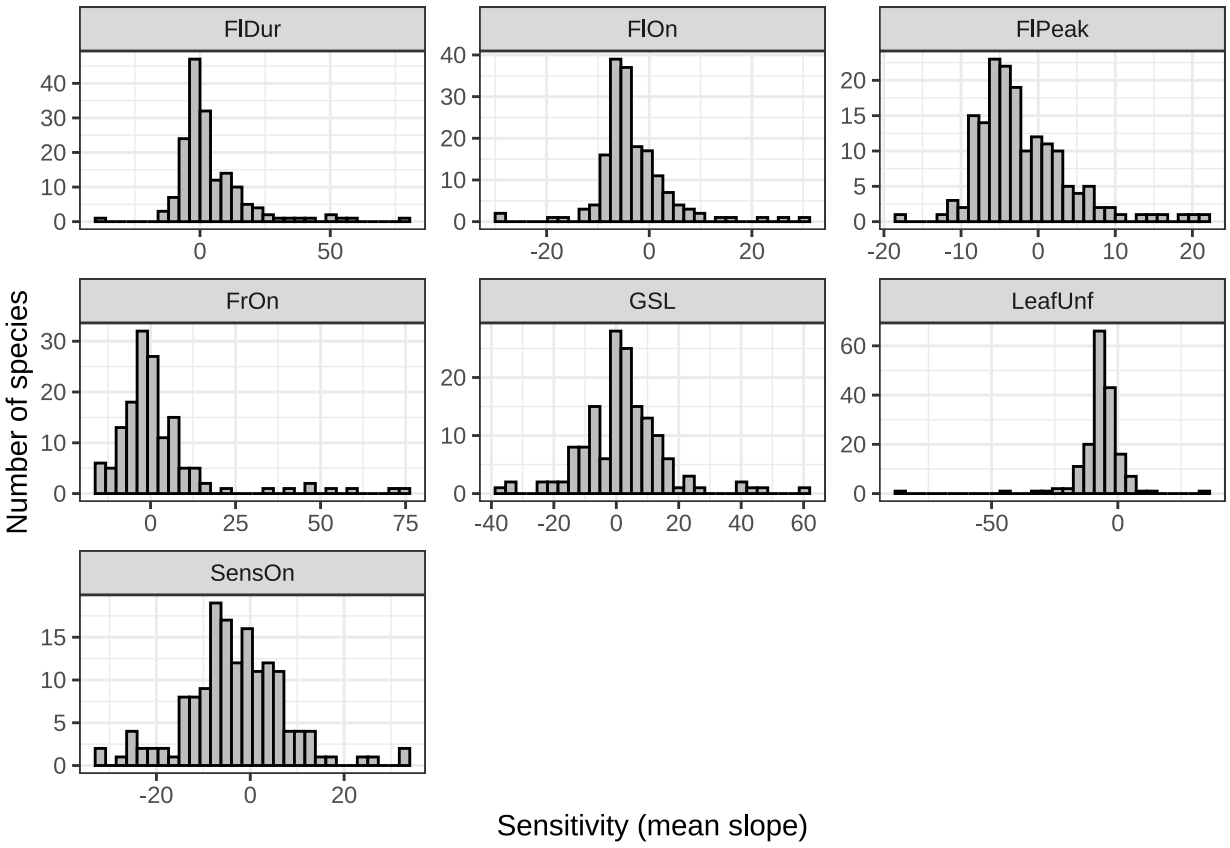

```
ggplot(sens_endanger, aes(x = factor(sens_01))) +
  geom_bar() +
  facet_wrap(~ stage) +
  labs(x = "Sensitive (0 = No, 1 = Yes)", y = "Number of species") +
  theme_bw()
```




```
ggplot(sens_endanger, aes(x = mean_slope)) +
  geom_histogram(color = "black", fill = "grey") +
  facet_wrap(~ stage, scales = "free") +
  labs(x = "Sensitivity (mean slope)", y = "Number of species") +
  theme_bw()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
## Warning: Removed 164 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



Models endangerment ~ sensitivity

Filtering slope_var < 50 - clearer patterns.

Added prediction plots only for models with any significant effects.

Mean slopes

LeafUnf, FLOn, FlPeak, FrOn, SensOn

```
mod_sens_neg <- sens_endanger %>%
  filter(slope_var < 50 & mean_slope < 0) %>%
  filter(!stage %in% c("FlDur", "GSL")) %>%
  group_nest(stage) %>%
  mutate(
    model = purrr::map(data,
                       ~ glmmTMB(cbind(threatened_countries,
                                       total_countries - threatened_countries) ~
                                   mean_slope,
                                 data = .x, family = betabinomial())))
mod_sens_neg_range <- sens_endanger %>%
  filter(slope_var < 50 & mean_slope < 0) %>%
```

```
  filter(!stage %in% c("FlDur", "GSL")) %>%
  group_nest(stage) %>%
  mutate(
    model = purrr::map(data,
                       ~ glmmTMB(prop_endangered_range ~ mean_slope,
                                 data = .x, family = beta_family(),
                                 ziformula=~.)))
mod_sens_neg_country_threat_ix <- sens_endanger %>%
  filter(slope_var < 50 & mean_slope < 0) %>%
  filter(!stage %in% c("FlDur", "GSL")) %>%
  group_nest(stage) %>%
  mutate(
    model = purrr::map(data,
                       ~ glmmTMB(country_threat_index ~ mean_slope,
                                 data = .x, family = beta_family(),
                                 ziformula=~.)))
mod_sens_neg_range_threat_ix <- sens_endanger %>%
  filter(slope_var < 50 & mean_slope < 0) %>%
  filter(!stage %in% c("FlDur", "GSL")) %>%
  group_nest(stage) %>%
  mutate(
    model = purrr::map(data,
                       ~ glmmTMB(range_threat_index ~ mean_slope,
                                 data = .x, family = beta_family(),
                                 ziformula=~.)))

mod_sens_neg %>% transmute(stage, results = map(model, tidy)) %>%
  unnest(results) %>% filter(term != "(Intercept)", p.value < 0.1)
```

Negative slopes only

```
## # A tibble: 2 x 8
##   stage   effect component term       estimate std.error statistic p.value
##   <chr>   <chr>  <chr>     <chr>         <dbl>     <dbl>     <dbl>   <dbl>
## 1 FlOn    fixed  cond      mean_slope   0.0791    0.0357      2.21  0.0269
## 2 LeafUnf fixed  cond      mean_slope  -0.0450    0.0246     -1.83  0.0670
```

```
mod_sens_neg_range %>% transmute(stage, results = map(model, tidy)) %>%
  unnest(results) %>% filter(term != "(Intercept)", p.value < 0.1)
```

```
## # A tibble: 1 x 8
##   stage  effect component term       estimate std.error statistic p.value
##   <chr>  <chr>  <chr>     <chr>         <dbl>     <dbl>     <dbl>   <dbl>
## 1 FlPeak fixed  zi        mean_slope   -0.142    0.0701     -2.02  0.0435
```

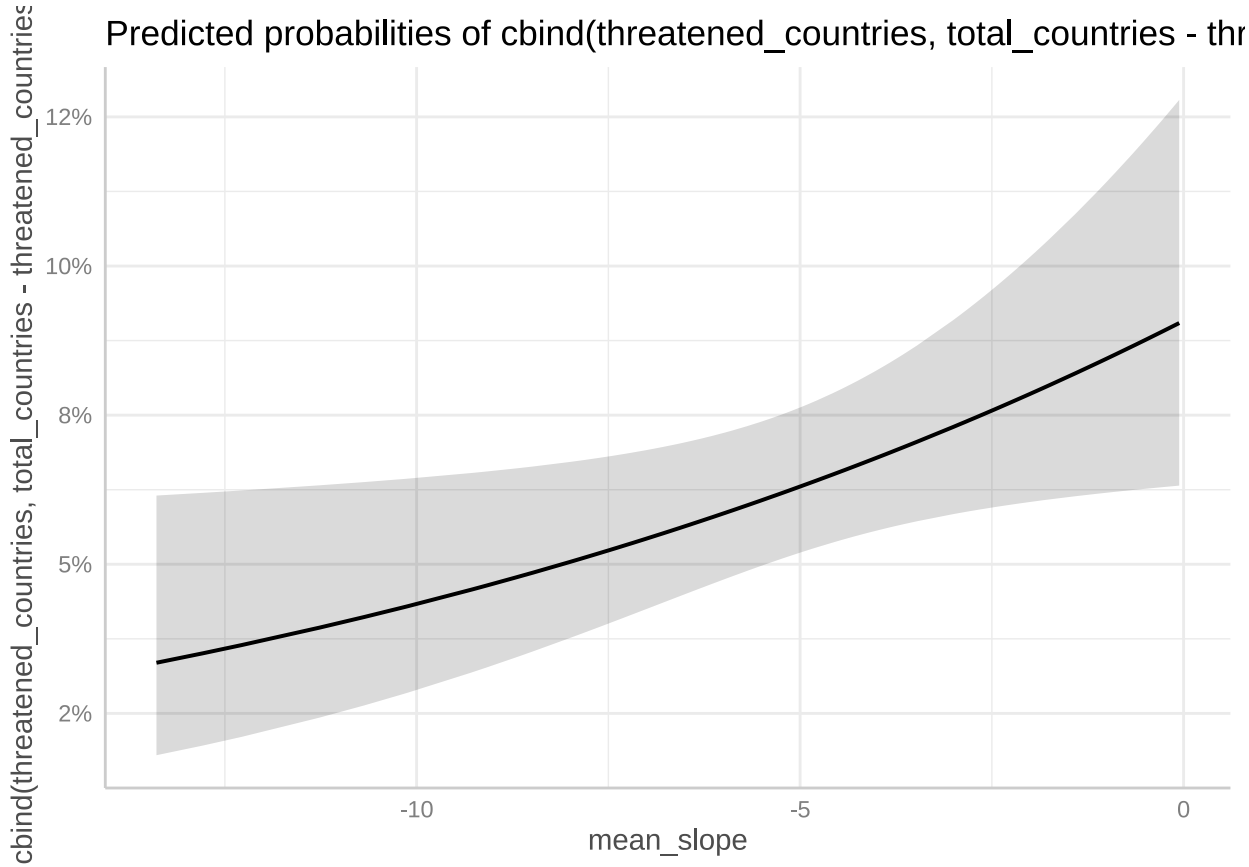
```
mod_sens_neg_country_threat_ix %>% transmute(stage, results = map(model, tidy)) %>%
  unnest(results) %>% filter(term != "(Intercept)", p.value < 0.1)
```

```
## # A tibble: 2 x 8
##   stage  effect component term       estimate std.error statistic  p.value
##   <chr>  <chr>  <chr>     <chr>         <dbl>     <dbl>     <dbl>    <dbl>
## 1 FlOn   fixed  cond      mean_slope   0.0861    0.0251      3.43 0.000595
## 2 FlPeak fixed  zi        mean_slope  -0.142     0.0701     -2.02 0.0435
```

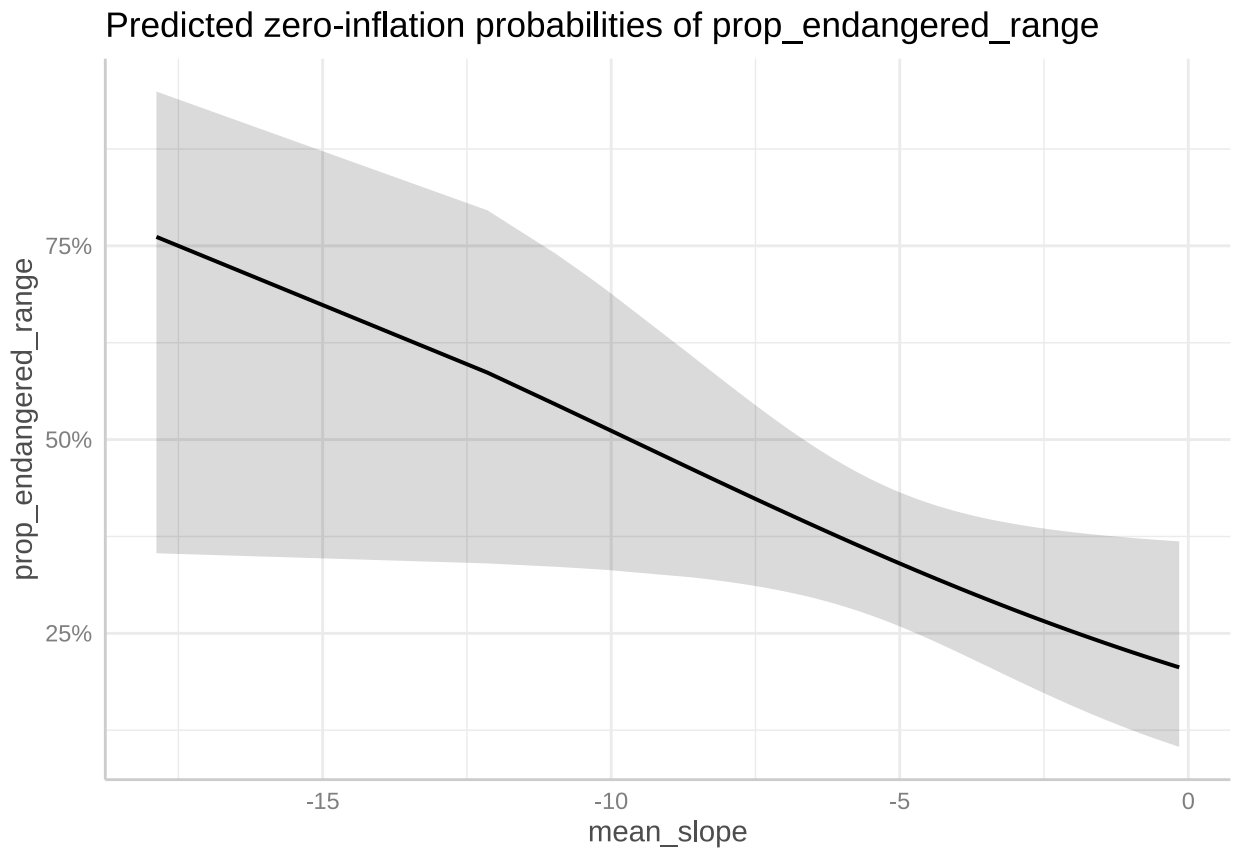
```
mod_sens_neg_range_threat_ix %>% transmute(stage, results = map(model, tidy)) %>%
  unnest(results) %>% filter(term != "(Intercept)", p.value < 0.1)
```

```
## # A tibble: 1 x 8
##   stage  effect component term       estimate std.error statistic p.value
##   <chr>  <chr>  <chr>     <chr>         <dbl>     <dbl>     <dbl>   <dbl>
## 1 FlPeak fixed  zi        mean_slope   -0.142    0.0701     -2.02  0.0435
```

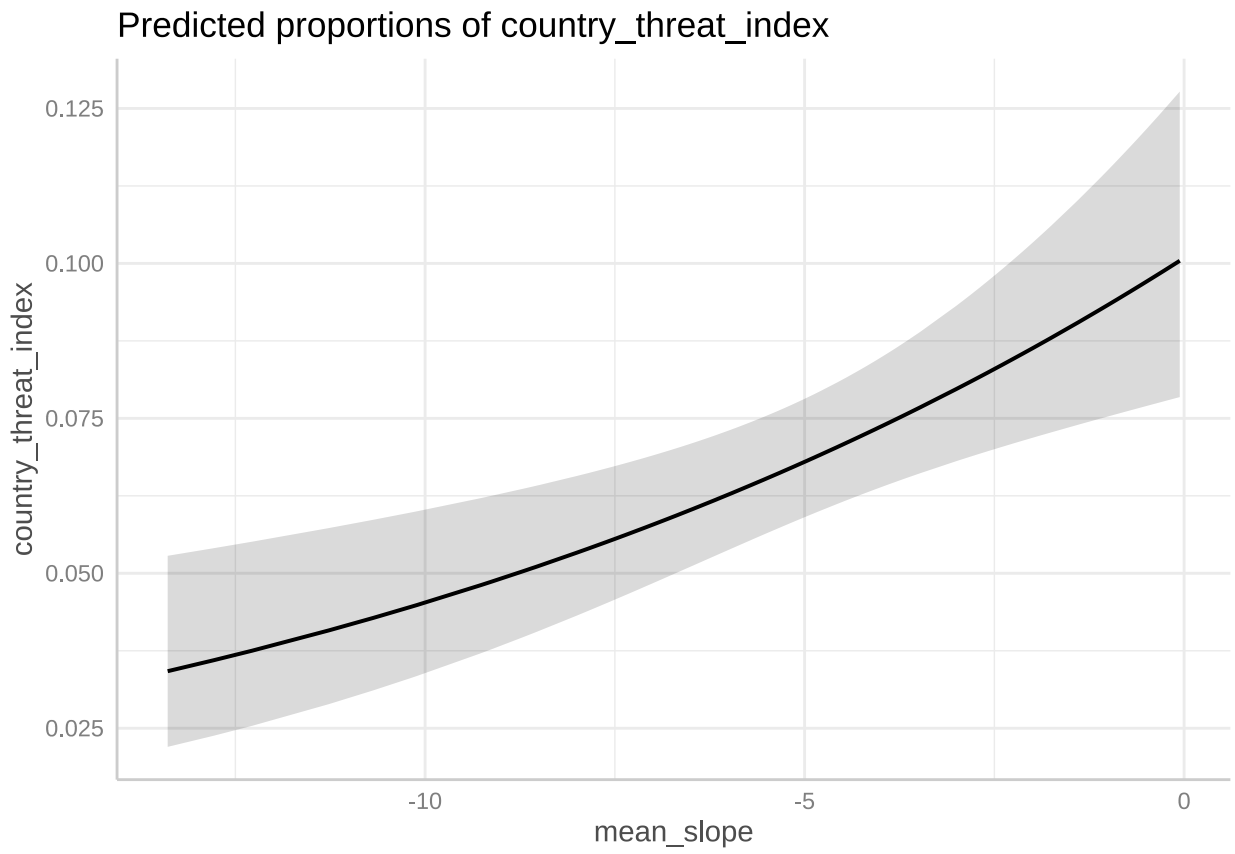
```
mod_sens_neg %>% filter(stage == "FlOn") %>% pull(model) %>% .[[1]] %>%
  ggpredict(terms = "mean_slope [all]") %>% plot() # p = 0.027
```



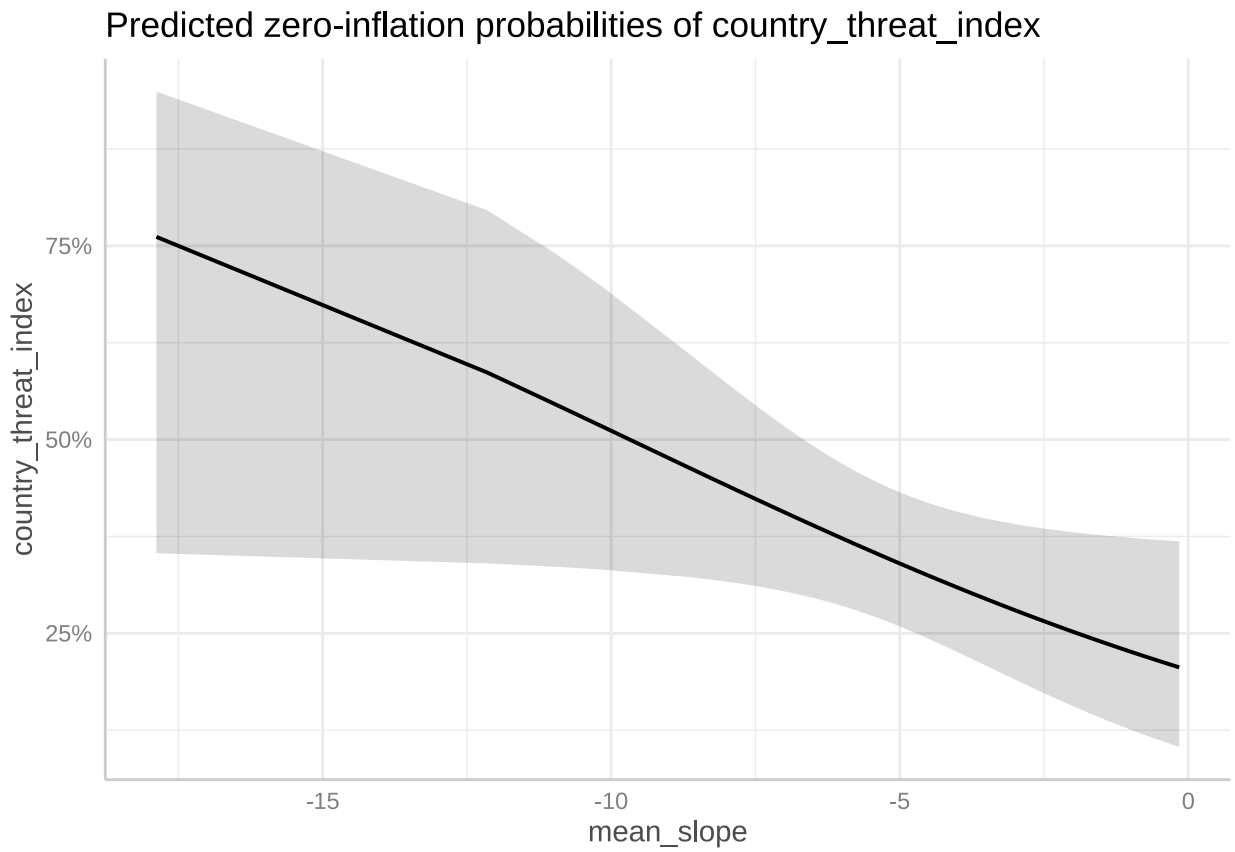

```
mod_sens_neg_range %>% filter(stage == "FlPeak") %>% pull(model) %>%
  .[[1]] %>%
  ggpredict(terms = "mean_slope [all]", type = "zi_prob") %>% plot() # p = 0.044
```



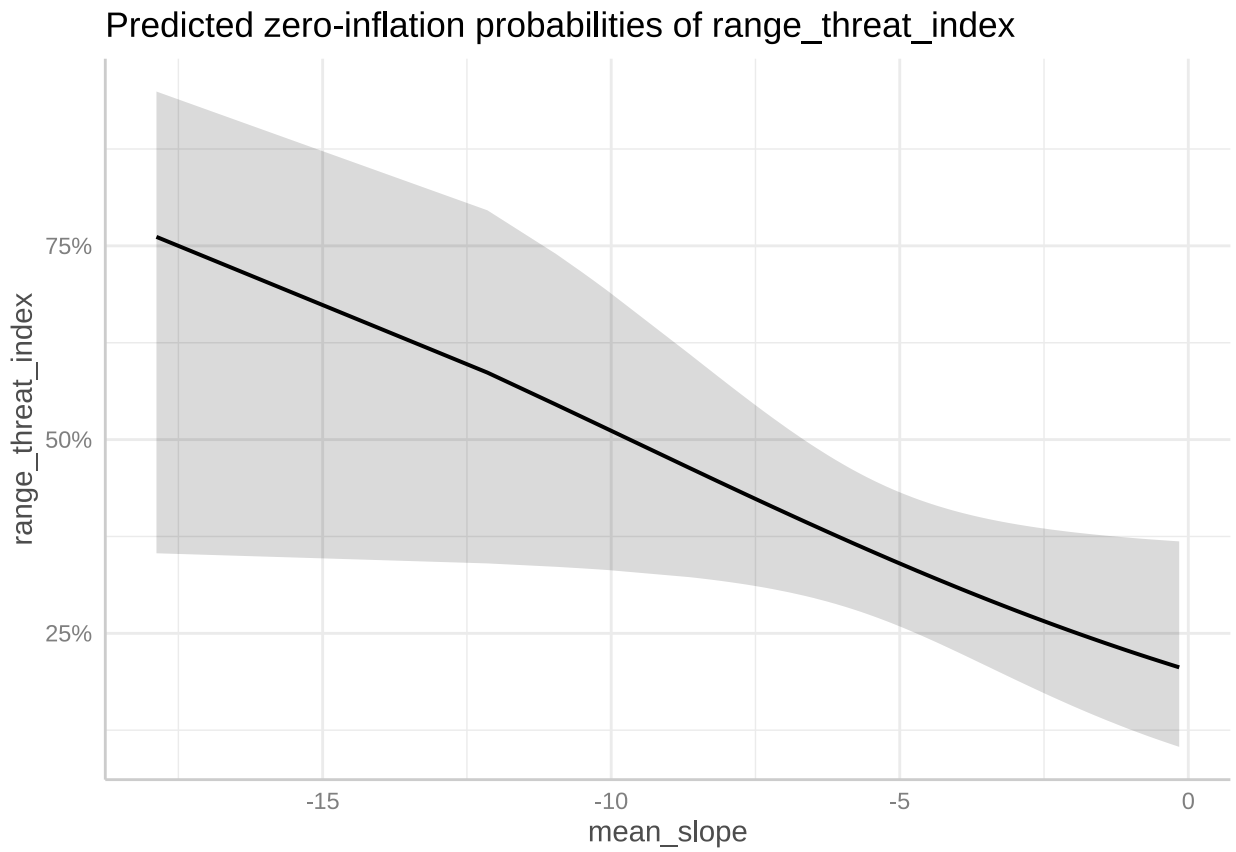

```
mod_sens_neg_country_threat_ix  %>% filter(stage == "FlOn") %>% pull(model) %>%
  .[[1]] %>%
  ggpredict(terms = "mean_slope [all]", type = "fixed") %>% plot() # p < 0.001
```




```
mod_sens_neg_country_threat_ix  %>% filter(stage == "FlPeak") %>%
  pull(model) %>% .[[1]] %>%
  ggpredict(terms = "mean_slope [all]", type = "zi_prob") %>% plot() # p = 0.044
```




```
mod_sens_neg_range_threat_ix  %>% filter(stage == "FlPeak") %>% pull(model) %>%
  .[[1]] %>%
  ggpredict(terms = "mean_slope [all]", type = "zi_prob") %>% plot() # p = 0.044 # zi part equal to pre
```




```
desired_order <- c("LeafUnf", "Fl0n", "FlPeak", "Fr0n", "Sens0n")

mods1 <- mod_sens_neg$model[
  match(desired_order, mod_sens_neg$stage)
]
mods2 <- mod_sens_neg_range$model[
  match(desired_order, mod_sens_neg_range$stage)
]
mods3 <- mod_sens_neg_country_threat_ix$model[
  match(desired_order, mod_sens_neg_country_threat_ix$stage)
]
mods4 <- mod_sens_neg_range_threat_ix$model[
  match(desired_order, mod_sens_neg_range_threat_ix$stage)
]

names(mods1) <- desired_order
names(mods2) <- desired_order
names(mods3) <- desired_order
names(mods4) <- desired_order

tab1 <- modelsummary(mods1, shape = model ~ term + statistic, output = "flextable",
                     coef_omit = "Intercept", fmt = 3,
                     estimate = c("Estimate" = "{estimate}"),
```

```
                    gof_omit = "AIC|BIC|RMSE|R2|Log.Lik|ICC|Num.Obs",
                    statistic = c("SE" = "{std.error}", "p-value" = "{p.value}"),
                    coef_rename = c(mean_slope = ""))
tab1 <- hline_top(tab1, border = fp_border(color = "white", width = 0),
                  part = "header") |>
  # Bold significant p-values
  bold(j = "p-value", i = ~ `p-value` < 0.05, bold = TRUE) |>
  font(fontname = "Times New Roman", part = "all")


tab2 <- modelsummary(mods2, shape = model + component ~ term + statistic,
                    output = "flextable",  coef_omit = "Intercept", fmt = 3,
                    estimate = c("Estimate" = "{estimate}"),
                    statistic = c("SE" = "{std.error}",
                                  "p-value" = "{p.value}"),
                    gof_omit = "AIC|BIC|RMSE|R2|Log.Lik|ICC|Num.Obs",
                    coef_rename = c(mean_slope = ""))
tab2 <- tab2$body$dataset |>
  mutate(component = recode(component, conditional = "Conditional",
                            zero_inflated = "Zero inflated")) |>
  flextable() |> set_header_labels(component = "Component") |>
  width(j = "component", width = 1.2) |>
  # Bold significant p-values
  bold(j = "p-value", i = ~ `p-value` < 0.05, bold = TRUE) |>
  hline_top(border = fp_border(color = "white", width = 0), part = "header") |>
  font(fontname = "Times New Roman", part = "all")

tab3 <- modelsummary(mods3, shape = model + component ~ term + statistic,
                    output = "flextable",  coef_omit = "Intercept", fmt = 3,
                    estimate = c("Estimate" = "{estimate}"),
                    statistic = c("SE" = "{std.error}",
                                  "p-value" = "{p.value}"),
                    gof_omit = "AIC|BIC|RMSE|R2|Log.Lik|ICC|Num.Obs",
                    coef_rename = c(mean_slope = ""))
tab3 <- tab3$body$dataset |>
  mutate(component = recode(component, conditional = "Conditional",
                            zero_inflated = "Zero inflated")) |>
  flextable() |> set_header_labels(component = "Component") |>
  width(j = "component", width = 1.2) |>
  # Bold significant p-values
  bold(j = "p-value", i = ~ `p-value` < 0.05, bold = TRUE) |>
  hline_top(border = fp_border(color = "white", width = 0), part = "header") |>
  font(fontname = "Times New Roman", part = "all")

tab4 <- modelsummary(mods4, shape = model + component ~ term + statistic,
                    output = "flextable",  coef_omit = "Intercept", fmt = 3,
                    estimate = c("Estimate" = "{estimate}"),
                    statistic = c("SE" = "{std.error}",
                                  "p-value" = "{p.value}"),
                    gof_omit = "AIC|BIC|RMSE|R2|Log.Lik|ICC|Num.Obs",
                    coef_rename = c(mean_slope = ""))
tab4 <- tab4$body$dataset |>
  mutate(component = recode(component, conditional = "Conditional",
```

```
                            zero_inflated = "Zero inflated")) |>
  flextable() |> set_header_labels(component = "Component") |>
  width(j = "component", width = 1.2) |>
  # Bold significant p-values
  bold(j = "p-value", i = ~ `p-value` < 0.05, bold = TRUE) |>
  hline_top(border = fp_border(color = "white", width = 0), part = "header") |>
  font(fontname = "Times New Roman", part = "all")
```

Model tables

```
mod_sens_pos <- sens_endanger %>%
  filter(slope_var < 50 & mean_slope > 0) %>%
  filter(!stage %in% c("FlDur", "GSL")) %>%
  group_nest(stage) %>%
  mutate(
    model = purrr::map(data,
                       ~ glmmTMB(cbind(threatened_countries,
                                       total_countries - threatened_countries) ~
                                   mean_slope,
                                 data = .x, family = betabinomial())))
mod_sens_pos_range <- sens_endanger %>%
  filter(slope_var < 50 & mean_slope > 0) %>%
  filter(!stage %in% c("FlDur", "GSL")) %>%
  group_nest(stage) %>%
  mutate(
    model = purrr::map(data,
                       ~ glmmTMB(prop_endangered_range ~ mean_slope,
                                 data = .x, family = beta_family(),
                                 ziformula=~.)))
mod_sens_pos_country_threat_ix <- sens_endanger %>%
  filter(slope_var < 50 & mean_slope > 0) %>%
  filter(!stage %in% c("FlDur", "GSL")) %>%
  group_nest(stage) %>%
  mutate(
    model = purrr::map(data,
                       ~ glmmTMB(country_threat_index ~ mean_slope,
                                 data = .x, family = beta_family(),
                                 ziformula=~.)))
mod_sens_pos_range_threat_ix <- sens_endanger %>%
  filter(slope_var < 50 & mean_slope > 0) %>%
  filter(!stage %in% c("FlDur", "GSL")) %>%
  group_nest(stage) %>%
  mutate(
    model = purrr::map(data,
                       ~ glmmTMB(range_threat_index ~ mean_slope,
                                 data = .x, family = beta_family(),
                                 ziformula=~.)))

mod_sens_pos %>% transmute(stage, results = map(model, tidy)) %>%
  unnest(results) %>% filter(term != "(Intercept)", p.value < 0.1)
```

Positive slopes only

```
## # A tibble: 0 x 8
## # i 8 variables: stage <chr>, effect <chr>, component <chr>, term <chr>,
## #   estimate <dbl>, std.error <dbl>, statistic <dbl>, p.value <dbl>
```

```
mod_sens_pos_range %>% transmute(stage, results = map(model, tidy)) %>%
  unnest(results) %>% filter(term != "(Intercept)", p.value < 0.1)
```

```
## # A tibble: 2 x 8
##   stage  effect component term       estimate std.error statistic p.value
##   <chr>  <chr>  <chr>     <chr>         <dbl>     <dbl>     <dbl>   <dbl>
## 1 FlPeak fixed  zi        mean_slope   0.161     0.0744      2.17  0.0300
## 2 FrOn   fixed  cond      mean_slope   0.0553    0.0263      2.10  0.0354
```

```
mod_sens_pos_country_threat_ix %>% transmute(stage, results = map(model, tidy)) %>%
  unnest(results) %>% filter(term != "(Intercept)", p.value < 0.1)
```

```
## # A tibble: 3 x 8
##   stage  effect component term       estimate std.error statistic   p.value
##   <chr>  <chr>  <chr>     <chr>         <dbl>     <dbl>     <dbl>     <dbl>
## 1 FlOn   fixed  cond      mean_slope    0.103    0.0448      2.30 0.0212
## 2 FlPeak fixed  cond      mean_slope    0.115    0.0287      4.03 0.0000565
## 3 FlPeak fixed  zi        mean_slope    0.161    0.0744      2.17 0.0300
```

```
mod_sens_pos_range_threat_ix %>% transmute(stage, results = map(model, tidy)) %>%
  unnest(results) %>% filter(term != "(Intercept)", p.value < 0.1)
```

```
## # A tibble: 2 x 8
##   stage  effect component term       estimate std.error statistic p.value
##   <chr>  <chr>  <chr>     <chr>         <dbl>     <dbl>     <dbl>   <dbl>
## 1 FlPeak fixed  zi        mean_slope   0.161     0.0744      2.17  0.0300
## 2 FrOn   fixed  cond      mean_slope   0.0584    0.0257      2.27  0.0230
```

```
mods5 <- mod_sens_pos$model[
  match(desired_order, mod_sens_pos$stage)
]
mods6 <- mod_sens_pos_range$model[
  match(desired_order, mod_sens_pos_range$stage)
]
mods7 <- mod_sens_pos_country_threat_ix$model[
  match(desired_order, mod_sens_pos_country_threat_ix$stage)
]
mods8 <- mod_sens_pos_range_threat_ix$model[
  match(desired_order, mod_sens_pos_range_threat_ix$stage)
]

names(mods5) <- desired_order
names(mods6) <- desired_order
```

```
names(mods7) <- desired_order
names(mods8) <- desired_order

tab5 <- modelsummary(mods5, shape = model ~ term + statistic, output = "flextable",
                     coef_omit = "Intercept", fmt = 3,
                     estimate = c("Estimate" = "{estimate}"),
                     gof_omit = "AIC|BIC|RMSE|R2|Log.Lik|ICC|Num.Obs",
                     statistic = c("SE" = "{std.error}", "p-value" = "{p.value}"),
                     coef_rename = c(mean_slope = ""))
tab5 <- hline_top(tab5, border = fp_border(color = "white", width = 0),
                  part = "header") |>
  # Bold significant p-values
  bold(j = "p-value", i = ~ `p-value` < 0.05, bold = TRUE) |>
  font(fontname = "Times New Roman", part = "all")


tab6 <- modelsummary(mods6, shape = model + component ~ term + statistic,
                     output = "flextable",  coef_omit = "Intercept", fmt = 3,
                     estimate = c("Estimate" = "{estimate}"),
                     statistic = c("SE" = "{std.error}",
                                   "p-value" = "{p.value}"),
                     gof_omit = "AIC|BIC|RMSE|R2|Log.Lik|ICC|Num.Obs",
                     coef_rename = c(mean_slope = ""))
tab6 <- tab6$body$dataset |>
  mutate(component = recode(component, conditional = "Conditional",
                            zero_inflated = "Zero inflated")) |>
  flextable() |> set_header_labels(component = "Component") |>
  width(j = "component", width = 1.2) |>
  # Bold significant p-values
  bold(j = "p-value", i = ~ `p-value` < 0.05, bold = TRUE) |>
  hline_top(border = fp_border(color = "white", width = 0), part = "header") |>
  font(fontname = "Times New Roman", part = "all")

tab7 <- modelsummary(mods7, shape = model + component ~ term + statistic,
                     output = "flextable",  coef_omit = "Intercept", fmt = 3,
                     estimate = c("Estimate" = "{estimate}"),
                     statistic = c("SE" = "{std.error}",
                                   "p-value" = "{p.value}"),
                     gof_omit = "AIC|BIC|RMSE|R2|Log.Lik|ICC|Num.Obs",
                     coef_rename = c(mean_slope = ""))
tab7 <- tab7$body$dataset |>
  mutate(component = recode(component, conditional = "Conditional",
                            zero_inflated = "Zero inflated")) |>
  flextable() |> set_header_labels(component = "Component") |>
  width(j = "component", width = 1.2) |>
  # Bold significant p-values
  bold(j = "p-value", i = ~ `p-value` < 0.05, bold = TRUE) |>
  hline_top(border = fp_border(color = "white", width = 0), part = "header") |>
  font(fontname = "Times New Roman", part = "all")

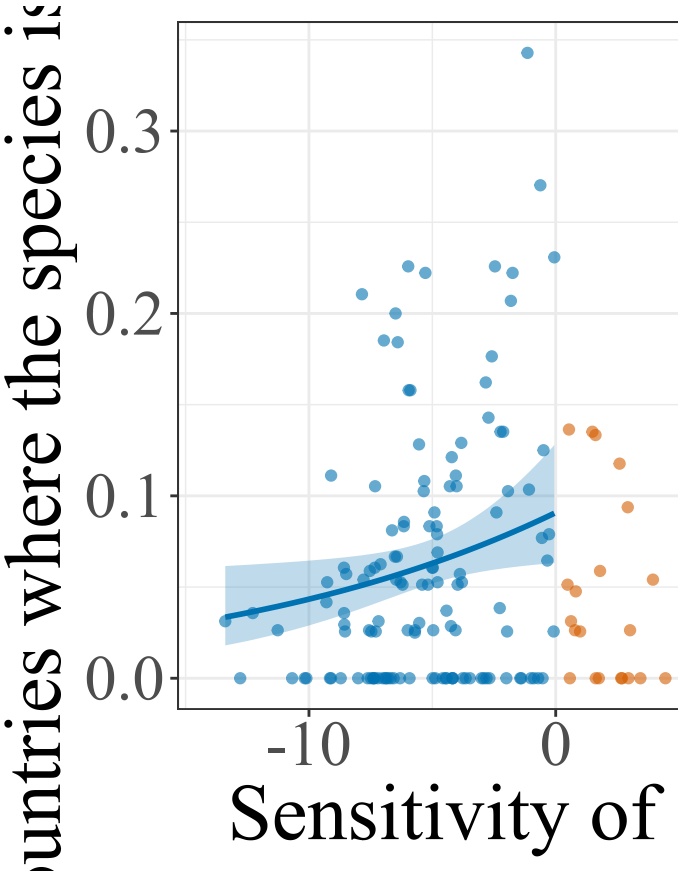
tab8 <- modelsummary(mods8, shape = model + component ~ term + statistic,
                     output = "flextable",  coef_omit = "Intercept", fmt = 3,
                     estimate = c("Estimate" = "{estimate}"),
```

```
                    statistic = c("SE" = "{std.error}",
                                  "p-value" = "{p.value}"),
                    gof_omit = "AIC|BIC|RMSE|R2|Log.Lik|ICC|Num.Obs",
                    coef_rename = c(mean_slope = ""))
tab8 <- tab8$body$dataset |>
  mutate(component = recode(component, conditional = "Conditional",
                            zero_inflated = "Zero inflated")) |>
  flextable() |> set_header_labels(component = "Component") |>
  width(j = "component", width = 1.2) |>
  # Bold significant p-values
  bold(j = "p-value", i = ~ `p-value` < 0.05, bold = TRUE) |>
  hline_top(border = fp_border(color = "white", width = 0), part = "header") |>
  font(fontname = "Times New Roman", part = "all")
```

Model tables

Figures

```
fig1 <- ggplot() +
    geom_ribbon(data = data.frame(
      mod_sens_neg %>% filter(stage == "FlOn") %>%
        pull(model) %>% .[[1]] %>%
        ggpredict(terms = "mean_slope [all]")),
      aes(x = x, y = predicted, ymin = conf.low, ymax = conf.high),
      alpha = 0.25, fill = "#0072B2") +
    geom_line(data = data.frame(
      mod_sens_neg %>% filter(stage == "FlOn") %>% pull(model) %>% .[[1]] %>%
        ggpredict(terms = "mean_slope [all]")
      ),
      aes(x = x, y = predicted), color = "#0072B2", linewidth = 1) +
  geom_point(data =
               sens_endanger %>%
               filter(slope_var < 50 & stage == "FlOn") %>%
               mutate(neg_pos = factor(ifelse(mean_slope< 0, "neg", "pos"))),
             aes(x = mean_slope, y = prop_endangerment, color = neg_pos), alpha = 0.6) +
  scale_color_manual(values = c(neg = "#0072B2", pos = "#D55E00")) +
  theme(legend.position = "none") +
  xlab("Sensitivity of flowering onset") +
  ylab("Prop. countries where the species is endangered")
fig1
```



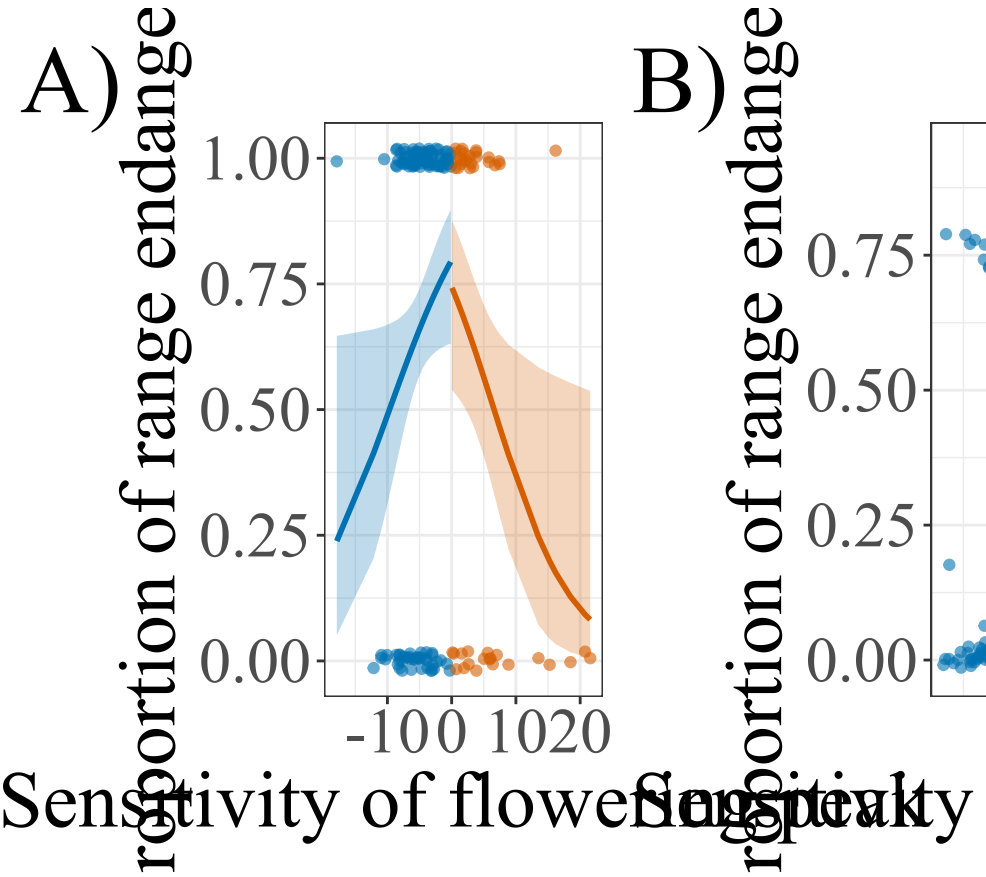
Proportion of countries where the species is endangered

```
fig2a <- ggplot() +
  # Negative
  geom_ribbon(data = data.frame(
    mod_sens_neg_range %>% filter(stage == "FlPeak") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "zi_prob")
    ),
    aes(x = x, y = 1-predicted, ymin = 1-conf.low, ymax = 1-conf.high),
    alpha = 0.25, fill = "#0072B2") +
  geom_line(data = data.frame(
    mod_sens_neg_range %>% filter(stage == "FlPeak") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "zi_prob")
    ),
    aes(x = x, y = 1-predicted), color = "#0072B2", linewidth = 1) +
  # Positive
  geom_ribbon(data = data.frame(
    mod_sens_pos_range %>% filter(stage == "FlPeak") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "zi_prob")),
    aes(x = x, y = 1 - predicted, ymin = 1-conf.low, ymax = 1-conf.high),
    alpha = 0.25, fill = "#D55E00") +
  geom_line(data = data.frame(
```

```
    mod_sens_pos_range %>% filter(stage == "FlPeak") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "zi_prob")),
    aes(x = x, y = 1 - predicted), color = "#D55E00", linewidth = 1) +
  # Points
  geom_jitter(data = sens_endanger %>%
                filter(slope_var < 50 & stage == "FlPeak") %>%
                 mutate(neg_pos = factor(ifelse(mean_slope< 0, "neg", "pos"))) %>%
                 mutate(y_01 = ifelse(prop_endangered_range == 0, 0, 1)),
              aes(x = mean_slope, y = y_01, color = neg_pos),
              alpha = 0.6, width = 0.02, height = 0.02) +
  scale_color_manual(values = c(neg = "#0072B2", pos = "#D55E00")) +
  theme(legend.position = "none") +
  xlab("Sensitivity of flowering peak") +
  ylab("Proportion of range endangered")

fig2b <- ggplot() +
  geom_ribbon(data = data.frame(
    mod_sens_pos_range %>% filter(stage == "FrOn") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "fixed")),
    aes(x = x, y = predicted, ymin = conf.low, ymax = conf.high),
    alpha = 0.25, fill = "#D55E00") +
  geom_line(data = data.frame(
    mod_sens_pos_range %>% filter(stage == "FrOn") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "fixed")),
    aes(x = x, y = predicted), color = "#D55E00", linewidth = 1) +
  geom_jitter(data = sens_endanger %>%
                filter(slope_var < 50 & stage == "FrOn") %>%
                 mutate(neg_pos = factor(ifelse(mean_slope< 0, "neg", "pos"))),
              aes(x = mean_slope, y = prop_endangered_range, color = neg_pos),
              alpha = 0.6, width = 0.02, height = 0.02) +
  scale_color_manual(values = c(neg = "#0072B2", pos = "#D55E00")) +
  theme(legend.position = "none") +
  xlab("Sensitivity of fruiting onset") +
  ylab("Proportion of range endangered")

fig2 <- fig2a + fig2b +
  plot_annotation(tag_levels = list(c("A)", "B)")))
fig2
```



Proportion of range endangered

```
fig3a <- ggplot() +
  # Negative
  geom_ribbon( data = data.frame(
    mod_sens_neg_country_threat_ix  %>% filter(stage == "FlOn") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "fixed")
    ),
    aes(x = x, y = predicted, ymin = conf.low, ymax = conf.high),
    alpha = 0.25, fill = "#0072B2") +
  geom_line( data = data.frame(
    mod_sens_neg_country_threat_ix  %>% filter(stage == "FlOn") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "fixed")
    ),
    aes(x = x, y = predicted), color = "#0072B2", linewidth = 1) +
  # Positive
  geom_ribbon(data = data.frame(
    mod_sens_pos_country_threat_ix %>% filter(stage == "FlOn") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "fixed")),
    aes(x = x, y = predicted, ymin = conf.low, ymax = conf.high),
    alpha = 0.25, fill = "#D55E00") +
  geom_line(data = data.frame(
```

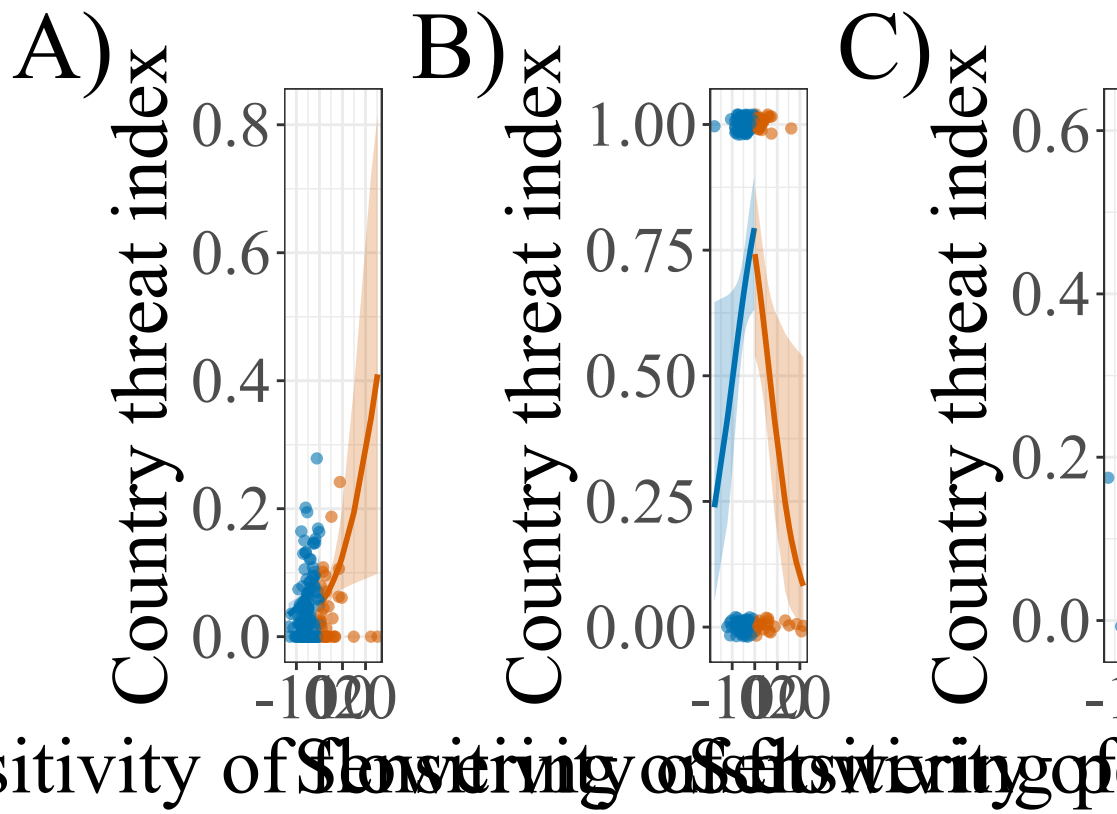
```
    mod_sens_pos_country_threat_ix %>% filter(stage == "FlOn") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "fixed")),
    aes(x = x, y = predicted), color = "#D55E00", linewidth = 1) +
  geom_point(data = sens_endanger %>%
               filter(slope_var < 50 & stage == "FlOn") %>%
               mutate(neg_pos = factor(ifelse(mean_slope< 0, "neg", "pos"))),
             aes(x = mean_slope, y = country_threat_index, color = neg_pos), alpha = 0.6) +
  scale_color_manual(values = c(neg = "#0072B2", pos = "#D55E00")) +
  theme(legend.position = "none") +
  xlab("Sensitivity of flowering onset") +
  ylab("Country threat index")

fig3b <- ggplot() +
  # Negative
  geom_ribbon( data = data.frame(
    mod_sens_neg_country_threat_ix  %>% filter(stage == "FlPeak") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "zi_prob")
    ),
    aes(x = x, y = 1 - predicted, ymin = 1 - conf.low, ymax = 1 - conf.high),
    alpha = 0.25, fill = "#0072B2") +
  geom_line( data = data.frame(
    mod_sens_neg_country_threat_ix  %>% filter(stage == "FlPeak") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "zi_prob")
    ),
    aes(x = x, y = 1 - predicted), color = "#0072B2", linewidth = 1) +
  # Positive
  geom_ribbon(data = data.frame(
    mod_sens_pos_country_threat_ix %>% filter(stage == "FlPeak") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "zi_prob")),
    aes(x = x, y = 1 - predicted, ymin = 1-conf.low, ymax = 1-conf.high),
    alpha = 0.25, fill = "#D55E00") +
  geom_line(data = data.frame(
    mod_sens_pos_country_threat_ix %>% filter(stage == "FlPeak") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "zi_prob")),
    aes(x = x, y = 1 - predicted), color = "#D55E00", linewidth = 1) +
  # Points
  geom_jitter(data = sens_endanger %>%
               filter(slope_var < 50 & stage == "FlPeak") %>%
                mutate(neg_pos = factor(ifelse(mean_slope< 0, "neg", "pos"))) %>%
               mutate(y_01 = ifelse(country_threat_index == 0, 0, 1)),
             aes(x = mean_slope, y = y_01, color = neg_pos),
             alpha = 0.6, width = 0.02, height = 0.02) +
  scale_color_manual(values = c(neg = "#0072B2", pos = "#D55E00")) +
  theme(legend.position = "none") +
  xlab("Sensitivity of flowering peak") +
  ylab("Country threat index")

fig3c <- ggplot() +
```

```
  geom_ribbon(data = data.frame(
    mod_sens_pos_country_threat_ix %>% filter(stage == "FlPeak") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "fixed")),
    aes(x = x, y = predicted, ymin = conf.low, ymax = conf.high),
    alpha = 0.25, fill = "#D55E00") +
  geom_line(data = data.frame(
    mod_sens_pos_country_threat_ix %>% filter(stage == "FlPeak") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "fixed")),
    aes(x = x, y = predicted), color = "#D55E00", linewidth = 1) +
  geom_jitter(data = sens_endanger %>%
                filter(slope_var < 50 & stage == "FlPeak") %>%
                 mutate(neg_pos = factor(ifelse(mean_slope< 0, "neg", "pos"))),
              aes(x = mean_slope, y = country_threat_index, color = neg_pos),
              alpha = 0.6, width = 0.02, height = 0.02) +
  scale_color_manual(values = c(neg = "#0072B2", pos = "#D55E00")) +
  theme(legend.position = "none") +
  xlab("Sensitivity of flowering peak") +
  ylab("Country threat index")

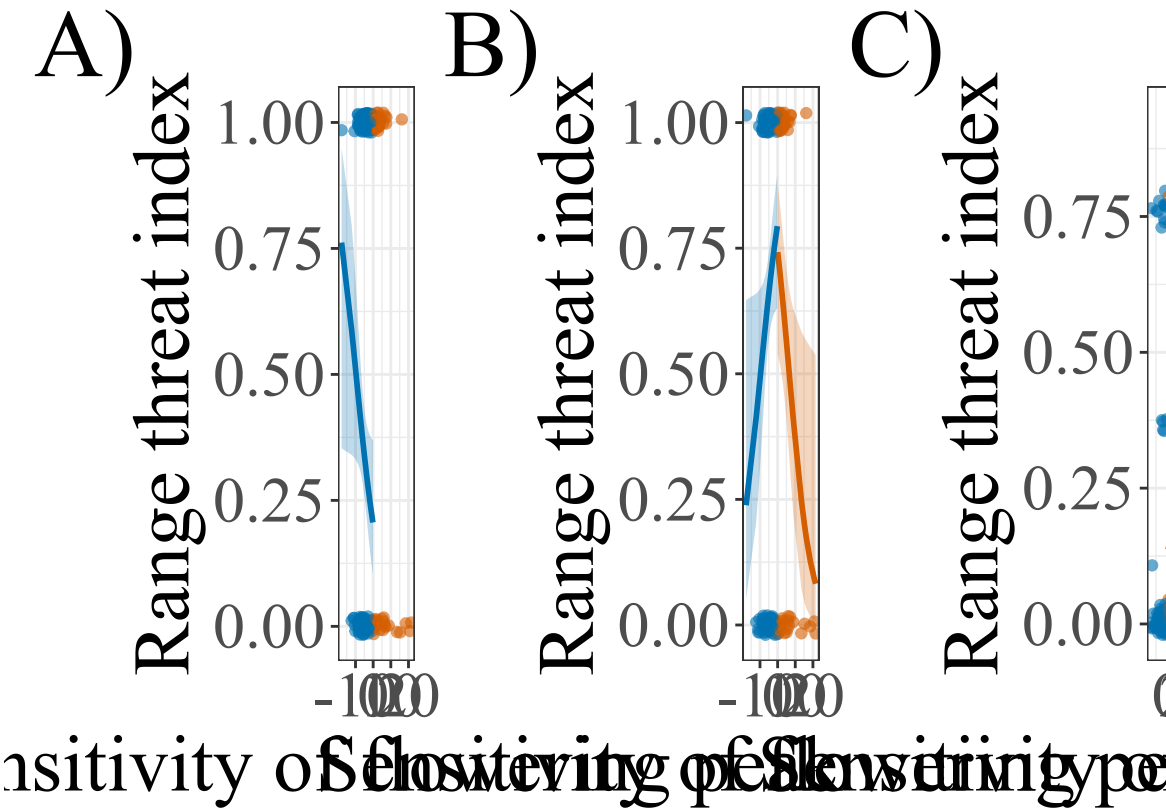
fig3 <- fig3a + fig3b + fig3c +
  plot_annotation(tag_levels = list(c("A)", "B)", "C)")))
fig3
```



Country threat index

```r
fig4a <- ggplot() +
  geom_ribbon( data = data.frame(
    mod_sens_neg_range_threat_ix  %>% filter(stage == "FlPeak") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "zi_prob")
    ),
    aes(x = x, y = predicted, ymin = conf.low, ymax = conf.high),
    alpha = 0.25, fill = "#0072B2") +
  geom_line( data = data.frame(
    mod_sens_neg_range_threat_ix  %>% filter(stage == "FlPeak") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "zi_prob")
    ),
    aes(x = x, y = predicted), color = "#0072B2", linewidth = 1) +
  geom_jitter(data = sens_endanger %>%
                filter(slope_var < 50 & stage == "FlPeak") %>%
                 mutate(neg_pos = factor(ifelse(mean_slope< 0, "neg", "pos"))) %>%
                mutate(y_01 = ifelse(range_threat_index == 0, 0, 1)),
              aes(x = mean_slope, y = y_01, color = neg_pos),
              alpha = 0.6, width = 0.02, height = 0.02) +
  scale_color_manual(values = c(neg = "#0072B2", pos = "#D55E00")) +
  theme(legend.position = "none") +
  xlab("Sensitivity of flowering peak") +
  ylab("Range threat index")
fig4b <- ggplot() +
  # Positive
  geom_ribbon(data = data.frame(
    mod_sens_pos_range_threat_ix %>% filter(stage == "FlPeak") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "zi_prob")),
    aes(x = x, y = 1 - predicted, ymin = 1-conf.low, ymax = 1-conf.high),
    alpha = 0.25, fill = "#D55E00") +
  geom_line(data = data.frame(
    mod_sens_pos_range_threat_ix %>% filter(stage == "FlPeak") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "zi_prob")),
    aes(x = x, y = 1 - predicted), color = "#D55E00", linewidth = 1) +
  # Negative
  geom_ribbon(data = data.frame(
    mod_sens_neg_range_threat_ix %>% filter(stage == "FlPeak") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "zi_prob")),
    aes(x = x, y = 1 - predicted, ymin = 1-conf.low, ymax = 1-conf.high),
    alpha = 0.25, fill = "#0072B2") +
  geom_line(data = data.frame(
    mod_sens_neg_range_threat_ix %>% filter(stage == "FlPeak") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "zi_prob")),
    aes(x = x, y = 1 - predicted), color = "#0072B2", linewidth = 1) +
  geom_jitter(data = sens_endanger %>%
                filter(slope_var < 50 & stage == "FlPeak") %>%
                 mutate(neg_pos = factor(ifelse(mean_slope< 0, "neg", "pos"))) %>%
```

```
                mutate(y_01 = ifelse(range_threat_index == 0, 0, 1)),
              aes(x = mean_slope, y = y_01, color = neg_pos),
              alpha = 0.6, width = 0.02, height = 0.02) +
  scale_color_manual(values = c(neg = "#0072B2", pos = "#D55E00")) +
  theme(legend.position = "none") +
  xlab("Sensitivity of flowering peak") +
  ylab("Range threat index")
fig4c <- ggplot() +
  geom_ribbon(data = data.frame(
    mod_sens_pos_range_threat_ix %>% filter(stage == "FrOn") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "fixed")),
    aes(x = x, y = predicted, ymin = conf.low, ymax = conf.high),
    alpha = 0.25, fill = "#D55E00") +
  geom_line(data = data.frame(
    mod_sens_pos_range_threat_ix %>% filter(stage == "FrOn") %>%
      pull(model) %>% .[[1]] %>%
      ggpredict(terms = "mean_slope [all]", type = "fixed")),
    aes(x = x, y = predicted), color = "#D55E00", linewidth = 1) +
  geom_jitter(data = sens_endanger %>%
                filter(slope_var < 50 & stage == "FrOn") %>%
                 mutate(neg_pos = factor(ifelse(mean_slope< 0, "neg", "pos"))),
              aes(x = mean_slope, y = range_threat_index, color = neg_pos),
              alpha = 0.6, width = 0.02, height = 0.02) +
  scale_color_manual(values = c(neg = "#0072B2", pos = "#D55E00")) +
  theme(legend.position = "none") +
  xlab("Sensitivity of fruting onset") +
  ylab("Range threat index")
fig4 <- fig4a + fig4b + fig4c +
  plot_annotation(tag_levels = list(c("A)", "B)", "C)")))
fig4
```



Range threat index

```
fig1 <- bundle_ggplot(fig1, width = 6, height = 6, units = "cm", scaling = "fixed")
fig2 <- bundle_ggplot(fig2, width = 12, height = 6, units = "cm", scaling = "fixed")
fig3 <- bundle_ggplot(fig3, width = 18, height = 6, units = "cm", scaling = "autoheight")
fig4 <- bundle_ggplot(fig4, width = 18, height = 6, units = "cm", scaling = "autoheight")
```

Prepare for kintted doc

FlDur and GSL: all values

```
# Models with quadratic term of slope
mod_sens_all_quad <- sens_endanger %>%
  filter(slope_var < 50) %>%
  filter(stage %in% c("FlDur", "GSL")) %>%
  group_nest(stage) %>%
  mutate(
    model = purrr::map(data,
                       ~ glmmTMB(cbind(threatened_countries,
                                       total_countries - threatened_countries) ~
                                   mean_slope + I(mean_slope^2),
                                 data = .x, family = betabinomial())))
mod_sens_all_range_quad <- sens_endanger %>%
```

```
  filter(slope_var < 50) %>%
  filter(stage %in% c("FlDur", "GSL")) %>%
  group_nest(stage) %>%
  mutate(
    model = purrr::map(data,
                       ~ glmmTMB(prop_endangered_range ~
                                   mean_slope + I(mean_slope^2),
                                 data = .x, family = beta_family(),
                                 ziformula=~.)))
mod_sens_all_country_threat_ix_quad <- sens_endanger %>%
  filter(slope_var < 50) %>%
  filter(stage %in% c("FlDur", "GSL")) %>%
  group_nest(stage) %>%
  mutate(
    model = purrr::map(data,
                       ~ glmmTMB(country_threat_index ~
                                   mean_slope + I(mean_slope^2),
                                 data = .x, family = beta_family(),
                                 ziformula=~.)))
mod_sens_all_range_threat_ix_quad <- sens_endanger %>%
  filter(slope_var < 50) %>%
  filter(stage %in% c("FlDur", "GSL")) %>%
  group_nest(stage) %>%
  mutate(
    model = purrr::map(data,
                       ~ glmmTMB(range_threat_index ~
                                   mean_slope + I(mean_slope^2),
                                 data = .x, family = beta_family(),
                                 ziformula=~.)))

# Models without quadratic term of slope
mod_sens_all <- sens_endanger %>%
  filter(slope_var < 50) %>%
  filter(stage %in% c("FlDur", "GSL")) %>%
  group_nest(stage) %>%
  mutate(
    model = purrr::map(data,
                       ~ glmmTMB(cbind(threatened_countries,
                                       total_countries - threatened_countries) ~
                                   mean_slope,
                                 data = .x, family = betabinomial())))
mod_sens_all_range <- sens_endanger %>%
  filter(slope_var < 50) %>%
  filter(stage %in% c("FlDur", "GSL")) %>%
  group_nest(stage) %>%
  mutate(
    model = purrr::map(data,
                       ~ glmmTMB(prop_endangered_range ~
                                   mean_slope,
                                 data = .x, family = beta_family(),
                                 ziformula=~.)))
mod_sens_all_country_threat_ix <- sens_endanger %>%
  filter(slope_var < 50) %>%
```

```
  filter(stage %in% c("FlDur", "GSL")) %>%
  group_nest(stage) %>%
  mutate(
    model = purrr::map(data,
                       ~ glmmTMB(country_threat_index ~
                                   mean_slope,
                                 data = .x, family = beta_family(),
                                 ziformula=~.)))
mod_sens_all_range_threat_ix <- sens_endanger %>%
  filter(slope_var < 50) %>%
  filter(stage %in% c("FlDur", "GSL")) %>%
  group_nest(stage) %>%
  mutate(
    model = purrr::map(data,
                       ~ glmmTMB(range_threat_index ~
                                   mean_slope,
                                 data = .x, family = beta_family(),
                                 ziformula=~.)))

mod_sens_all_quad %>% transmute(stage, results = map(model, tidy)) %>%
  unnest(results) %>% filter(term != "(Intercept)", p.value < 0.1)
```

```
## # A tibble: 0 x 8
## # i 8 variables: stage <chr>, effect <chr>, component <chr>, term <chr>,
## #   estimate <dbl>, std.error <dbl>, statistic <dbl>, p.value <dbl>
```

```
mod_sens_all_range_quad %>% transmute(stage, results = map(model, tidy)) %>%
  unnest(results) %>% filter(term != "(Intercept)", p.value < 0.1)
```

```
## # A tibble: 1 x 8
##   stage effect component term       estimate std.error statistic p.value
##   <chr> <chr>  <chr>     <chr>         <dbl>     <dbl>     <dbl>   <dbl>
## 1 GSL   fixed  cond      mean_slope   0.0546    0.0291      1.88  0.0602
```

```
mod_sens_all_country_threat_ix_quad %>% transmute(stage, results = map(model, tidy)) %>%
  unnest(results) %>% filter(term != "(Intercept)", p.value < 0.1)
```

```
## # A tibble: 0 x 8
## # i 8 variables: stage <chr>, effect <chr>, component <chr>, term <chr>,
## #   estimate <dbl>, std.error <dbl>, statistic <dbl>, p.value <dbl>
```

```
mod_sens_all_range_threat_ix_quad %>% transmute(stage, results = map(model, tidy)) %>%
  unnest(results) %>% filter(term != "(Intercept)", p.value < 0.1)
```

```
## # A tibble: 1 x 8
##   stage effect component term       estimate std.error statistic p.value
##   <chr> <chr>  <chr>     <chr>         <dbl>     <dbl>     <dbl>   <dbl>
## 1 GSL   fixed  cond      mean_slope   0.0481    0.0285      1.68  0.0921
```

```
mod_sens_all %>% transmute(stage, results = map(model, tidy)) %>%
  unnest(results) %>% filter(term != "(Intercept)", p.value < 0.1)
```

```
## # A tibble: 0 x 8
## # i 8 variables: stage <chr>, effect <chr>, component <chr>, term <chr>,
## #   estimate <dbl>, std.error <dbl>, statistic <dbl>, p.value <dbl>
```

```
mod_sens_all_range %>% transmute(stage, results = map(model, tidy)) %>%
  unnest(results) %>% filter(term != "(Intercept)", p.value < 0.1)
```

```
## # A tibble: 0 x 8
## # i 8 variables: stage <chr>, effect <chr>, component <chr>, term <chr>,
## #   estimate <dbl>, std.error <dbl>, statistic <dbl>, p.value <dbl>
```

```
mod_sens_all_country_threat_ix %>% transmute(stage, results = map(model, tidy)) %>%
  unnest(results) %>% filter(term != "(Intercept)", p.value < 0.1)
```

```
## # A tibble: 0 x 8
## # i 8 variables: stage <chr>, effect <chr>, component <chr>, term <chr>,
## #   estimate <dbl>, std.error <dbl>, statistic <dbl>, p.value <dbl>
```

```
mod_sens_all_range_threat_ix %>% transmute(stage, results = map(model, tidy)) %>%
  unnest(results) %>% filter(term != "(Intercept)", p.value < 0.1)
```

```
## # A tibble: 0 x 8
## # i 8 variables: stage <chr>, effect <chr>, component <chr>, term <chr>,
## #   estimate <dbl>, std.error <dbl>, statistic <dbl>, p.value <dbl>
```

Model tables

```
desired_order_2 <- c("FlDur", "GSL")

mods9<- mod_sens_all_quad$model[
  match(desired_order_2, mod_sens_all_quad$stage)
]
mods10 <- mod_sens_all_range_quad$model[
  match(desired_order_2, mod_sens_all_range_quad$stage)
]
mods11 <- mod_sens_all_country_threat_ix_quad$model[
  match(desired_order_2, mod_sens_all_country_threat_ix_quad$stage)
]
mods12 <- mod_sens_all_range_threat_ix_quad$model[
  match(desired_order_2, mod_sens_all_range_threat_ix_quad$stage)
]

names(mods9) <- desired_order_2
names(mods10) <- desired_order_2
```

```r
names(mods11) <- desired_order_2
names(mods12) <- desired_order_2

tab9 <- modelsummary(mods9, shape = model ~ term + statistic, output = "flextable",
                     coef_omit = "Intercept", fmt = 3,
                     estimate = c("Estimate" = "{estimate}"),
                     gof_omit = "AIC|BIC|RMSE|R2|Log.Lik|ICC|Num.Obs",
                     statistic = c("SE" = "{std.error}", "p-value" = "{p.value}"),
                     coef_rename = c(mean_slope = "Linear", "I(mean_slope^2)" = "Quadratic"))
tab9 <- hline_top(tab9, border = fp_border(color = "white", width = 0),
                  part = "header") |>
  # Bold significant p-values
  bold(j = "p-value", i = ~ `p-value` < 0.05, bold = TRUE) |>
  font(fontname = "Times New Roman", part = "all")


tab10 <- modelsummary(mods10, shape = model + component ~ term + statistic,
                     output = "flextable",  coef_omit = "Intercept", fmt = 3,
                     estimate = c("Estimate" = "{estimate}"),
                     statistic = c("SE" = "{std.error}",
                                   "p-value" = "{p.value}"),
                     gof_omit = "AIC|BIC|RMSE|R2|Log.Lik|ICC|Num.Obs",
                     coef_rename = c(mean_slope = "Linear", "I(mean_slope^2)" = "Quadratic"))
tab10 <- tab10$body$dataset |>
  mutate(component = recode(component, conditional = "Conditional",
                            zero_inflated = "Zero inflated")) |>
  flextable() |> set_header_labels(component = "Component") |>
  width(j = "component", width = 1.2) |>
  # Bold significant p-values
  bold(j = "p-value", i = ~ `p-value` < 0.05, bold = TRUE) |>
  hline_top(border = fp_border(color = "white", width = 0), part = "header") |>
  font(fontname = "Times New Roman", part = "all")

tab11 <- modelsummary(mods11, shape = model + component ~ term + statistic,
                     output = "flextable",  coef_omit = "Intercept", fmt = 3,
                     estimate = c("Estimate" = "{estimate}"),
                     statistic = c("SE" = "{std.error}",
                                   "p-value" = "{p.value}"),
                     gof_omit = "AIC|BIC|RMSE|R2|Log.Lik|ICC|Num.Obs",
                     coef_rename = c(mean_slope = "Linear", "I(mean_slope^2)" = "Quadratic"))
tab11 <- tab11$body$dataset |>
  mutate(component = recode(component, conditional = "Conditional",
                            zero_inflated = "Zero inflated")) |>
  flextable() |> set_header_labels(component = "Component") |>
  width(j = "component", width = 1.2) |>
  # Bold significant p-values
  bold(j = "p-value", i = ~ `p-value` < 0.05, bold = TRUE) |>
  hline_top(border = fp_border(color = "white", width = 0), part = "header") |>
  font(fontname = "Times New Roman", part = "all")

tab12 <- modelsummary(mods12, shape = model + component ~ term + statistic,
                     output = "flextable",  coef_omit = "Intercept", fmt = 3,
                     estimate = c("Estimate" = "{estimate}"),
```

```
                    statistic = c("SE" = "{std.error}",
                                  "p-value" = "{p.value}"),
                    gof_omit = "AIC|BIC|RMSE|R2|Log.Lik|ICC|Num.Obs",
                    coef_rename = c(mean_slope = "Linear", "I(mean_slope^2)" = "Quadratic"))
tab12 <- tab12$body$dataset |>
  mutate(component = recode(component, conditional = "Conditional",
                            zero_inflated = "Zero inflated")) |>
  flextable() |> set_header_labels(component = "Component") |>
  width(j = "component", width = 1.2) |>
  # Bold significant p-values
  bold(j = "p-value", i = ~ `p-value` < 0.05, bold = TRUE) |>
  hline_top(border = fp_border(color = "white", width = 0), part = "header") |>
  font(fontname = "Times New Roman", part = "all")
```

With quadratic term

```
mods13<- mod_sens_all$model[
  match(desired_order_2, mod_sens_all$stage)
]
mods14 <- mod_sens_all_range$model[
  match(desired_order_2, mod_sens_all_range$stage)
]
mods15 <- mod_sens_all_country_threat_ix$model[
  match(desired_order_2, mod_sens_all_country_threat_ix$stage)
]
mods16 <- mod_sens_all_range_threat_ix$model[
  match(desired_order_2, mod_sens_all_range_threat_ix$stage)
]

names(mods13) <- desired_order_2
names(mods14) <- desired_order_2
names(mods15) <- desired_order_2
names(mods16) <- desired_order_2

tab13 <- modelsummary(mods13, shape = model ~ term + statistic, output = "flextable",
                    coef_omit = "Intercept", fmt = 3,
                    estimate = c("Estimate" = "{estimate}"),
                    gof_omit = "AIC|BIC|RMSE|R2|Log.Lik|ICC|Num.Obs",
                    statistic = c("SE" = "{std.error}", "p-value" = "{p.value}"),
                    coef_rename = c(mean_slope = ""))
tab13 <- hline_top(tab13, border = fp_border(color = "white", width = 0),
                   part = "header") |>
  # Bold significant p-values
  bold(j = "p-value", i = ~ `p-value` < 0.05, bold = TRUE) |>
  font(fontname = "Times New Roman", part = "all")


tab14 <- modelsummary(mods14, shape = model + component ~ term + statistic,
                    output = "flextable",  coef_omit = "Intercept", fmt = 3,
                    estimate = c("Estimate" = "{estimate}"),
```

```
                    statistic = c("SE" = "{std.error}",
                                  "p-value" = "{p.value}"),
                    gof_omit = "AIC|BIC|RMSE|R2|Log.Lik|ICC|Num.Obs",
                    coef_rename = c(mean_slope = ""))
tab14 <- tab14$body$dataset |>
  mutate(component = recode(component, conditional = "Conditional",
                            zero_inflated = "Zero inflated")) |>
  flextable() |> set_header_labels(component = "Component") |>
  width(j = "component", width = 1.2) |>
  # Bold significant p-values
  bold(j = "p-value", i = ~ `p-value` < 0.05, bold = TRUE) |>
  hline_top(border = fp_border(color = "white", width = 0), part = "header") |>
  font(fontname = "Times New Roman", part = "all")

tab15 <- modelsummary(mods15, shape = model + component ~ term + statistic,
                    output = "flextable",  coef_omit = "Intercept", fmt = 3,
                    estimate = c("Estimate" = "{estimate}"),
                    statistic = c("SE" = "{std.error}",
                                  "p-value" = "{p.value}"),
                    gof_omit = "AIC|BIC|RMSE|R2|Log.Lik|ICC|Num.Obs",
                    coef_rename = c(mean_slope = ""))
tab15 <- tab15$body$dataset |>
  mutate(component = recode(component, conditional = "Conditional",
                            zero_inflated = "Zero inflated")) |>
  flextable() |> set_header_labels(component = "Component") |>
  width(j = "component", width = 1.2) |>
  # Bold significant p-values
  bold(j = "p-value", i = ~ `p-value` < 0.05, bold = TRUE) |>
  hline_top(border = fp_border(color = "white", width = 0), part = "header") |>
  font(fontname = "Times New Roman", part = "all")

tab16 <- modelsummary(mods16, shape = model + component ~ term + statistic,
                    output = "flextable",  coef_omit = "Intercept", fmt = 3,
                    estimate = c("Estimate" = "{estimate}"),
                    statistic = c("SE" = "{std.error}",
                                  "p-value" = "{p.value}"),
                    gof_omit = "AIC|BIC|RMSE|R2|Log.Lik|ICC|Num.Obs",
                    coef_rename = c(mean_slope = ""))
tab16 <- tab16$body$dataset |>
  mutate(component = recode(component, conditional = "Conditional",
                            zero_inflated = "Zero inflated")) |>
  flextable() |> set_header_labels(component = "Component") |>
  width(j = "component", width = 1.2) |>
  # Bold significant p-values
  bold(j = "p-value", i = ~ `p-value` < 0.05, bold = TRUE) |>
  hline_top(border = fp_border(color = "white", width = 0), part = "header") |>
  font(fontname = "Times New Roman", part = "all")
```

Without quadratic term

PCA

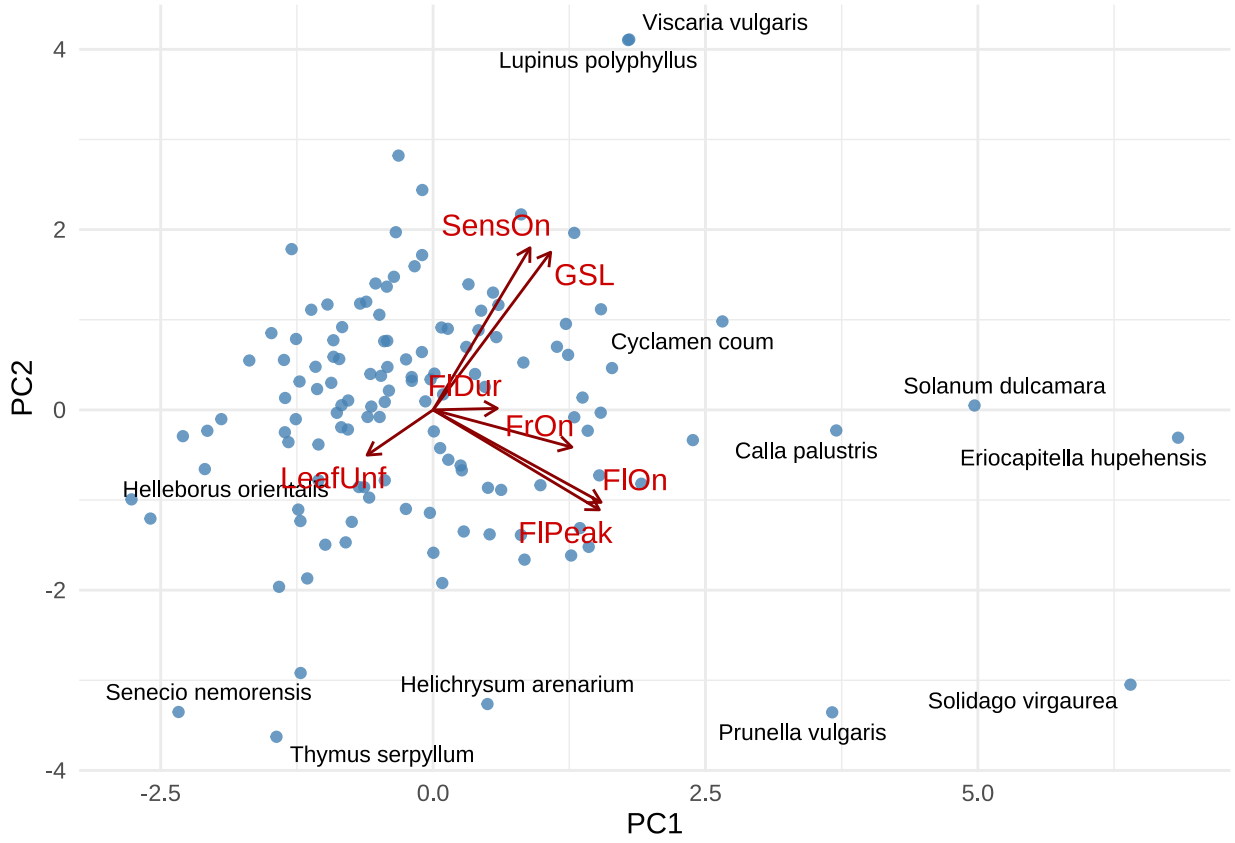
```
pca_phen <- prcomp(sens_per_endanger %>%
                     dplyr::select(starts_with("mean_slope_")) %>%
                     drop_na(),
                   center = TRUE, scale. = TRUE)
```

```
scores <- as.data.frame(pca_phen$x) %>%
  mutate(species = (sens_per_endanger %>%
                      dplyr::select(species, starts_with("mean_slope_")) %>%
                      drop_na())$species) %>%
  mutate(label = ifelse(abs(PC1) > quantile(abs(PC1), 0.95) |
                          abs(PC2) > quantile(abs(PC2), 0.95),
                        species, ""))
```

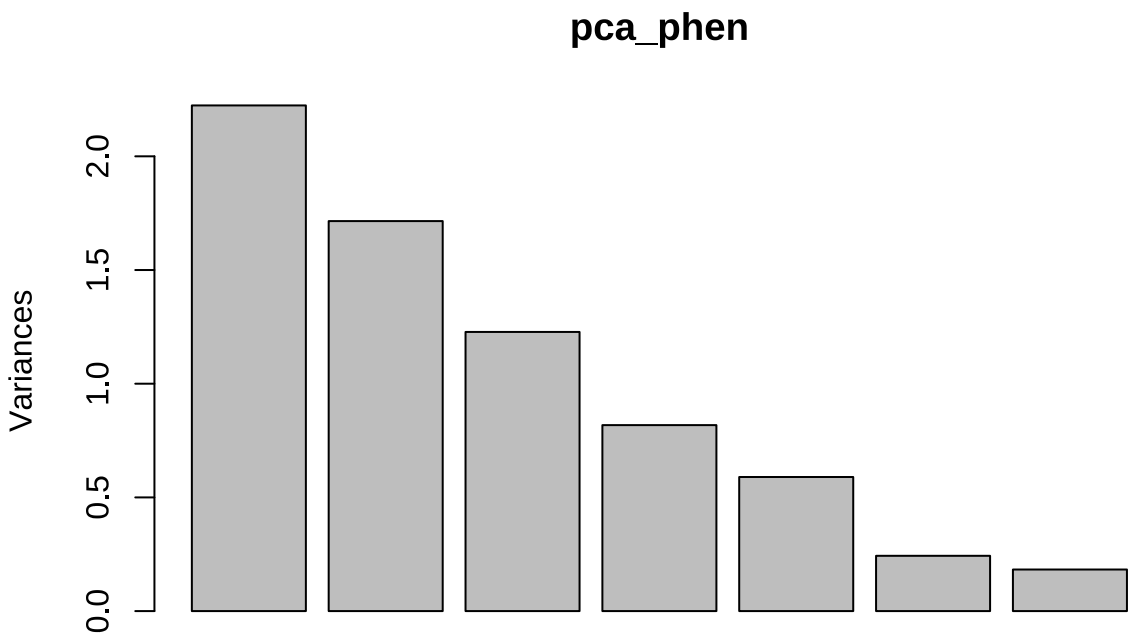
```
# Variable loadings for arrows
loadings <- as.data.frame(pca_phen$rotation) %>%
  mutate(variable = rownames(.)) %>%
  # Remove mean_slope from variable names
  mutate(variable = gsub("mean_slope_", "", variable))

ggplot(scores %>%
         left_join(
           sens_per_endanger %>%
             dplyr::select(species, prop_endangerment)
           ),
       aes(x = PC1, y = PC2)) +
  geom_point(alpha = 0.8, color = "steelblue"
             ) +
  geom_text_repel(aes(label = label), size = 3, max.overlaps = 15) +
  geom_segment(data = loadings,
               aes(x = 0, y = 0, xend = PC1*3, yend = PC2*3),
               arrow = arrow(length = unit(0.2, "cm")),
               color = "darkred") +
  geom_text_repel(data = loadings,
                  aes(x = PC1*3, y = PC2*3, label = variable),
                  color = "red3", size = 4) +
  theme_minimal() +
  labs(x = "PC1", y = "PC2")
```

```
## Joining with `by = join_by(species)`
```

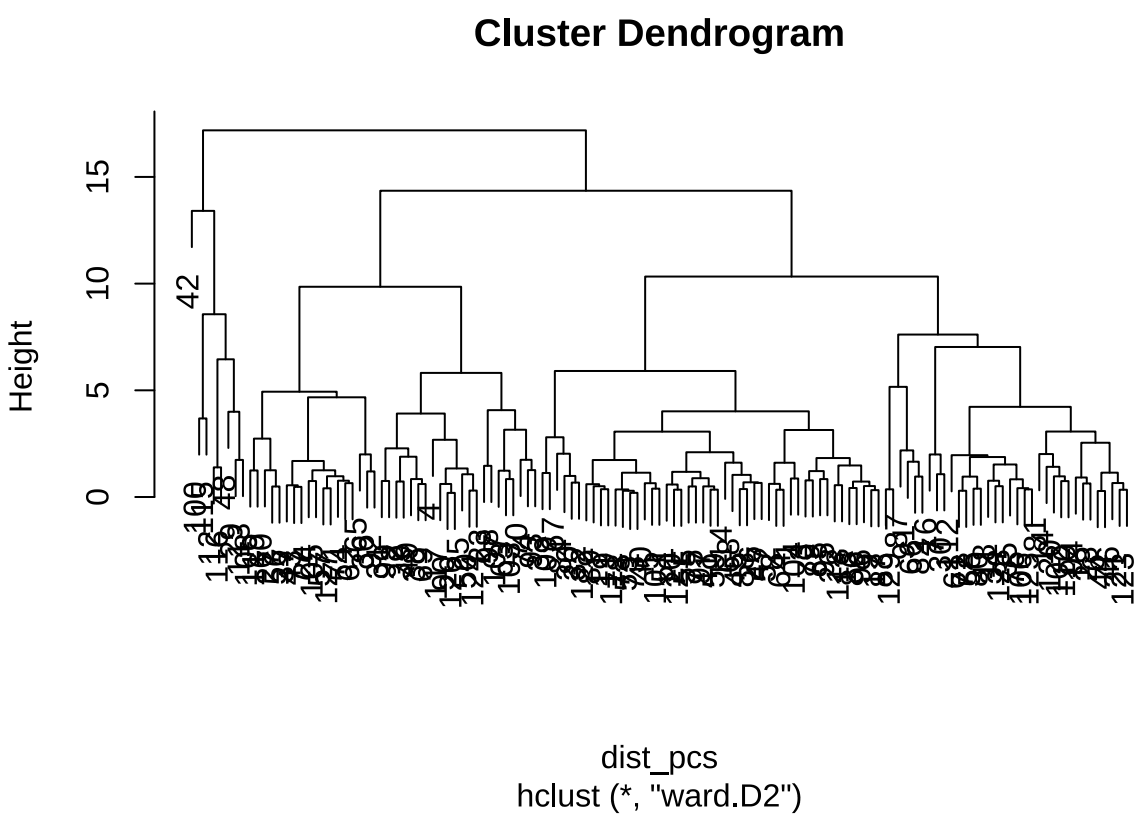



```
plot(pca_phen)
```

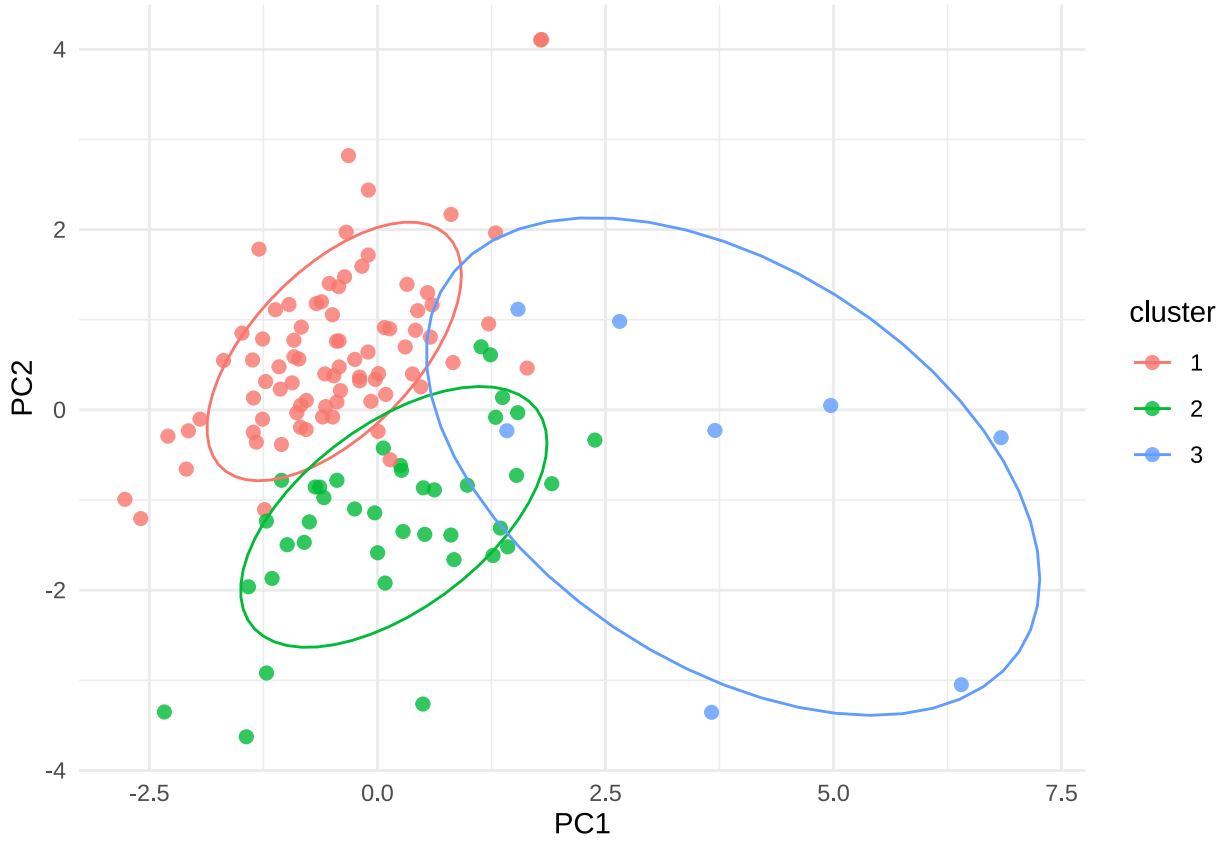


Clustering

```
dist_pcs <- dist(scores %>%
                   dplyr::select(PC1, PC2, PC3, PC4),
                 method = "euclidean")
hc <- hclust(dist_pcs, method = "ward.D2")
plot(hc)
```



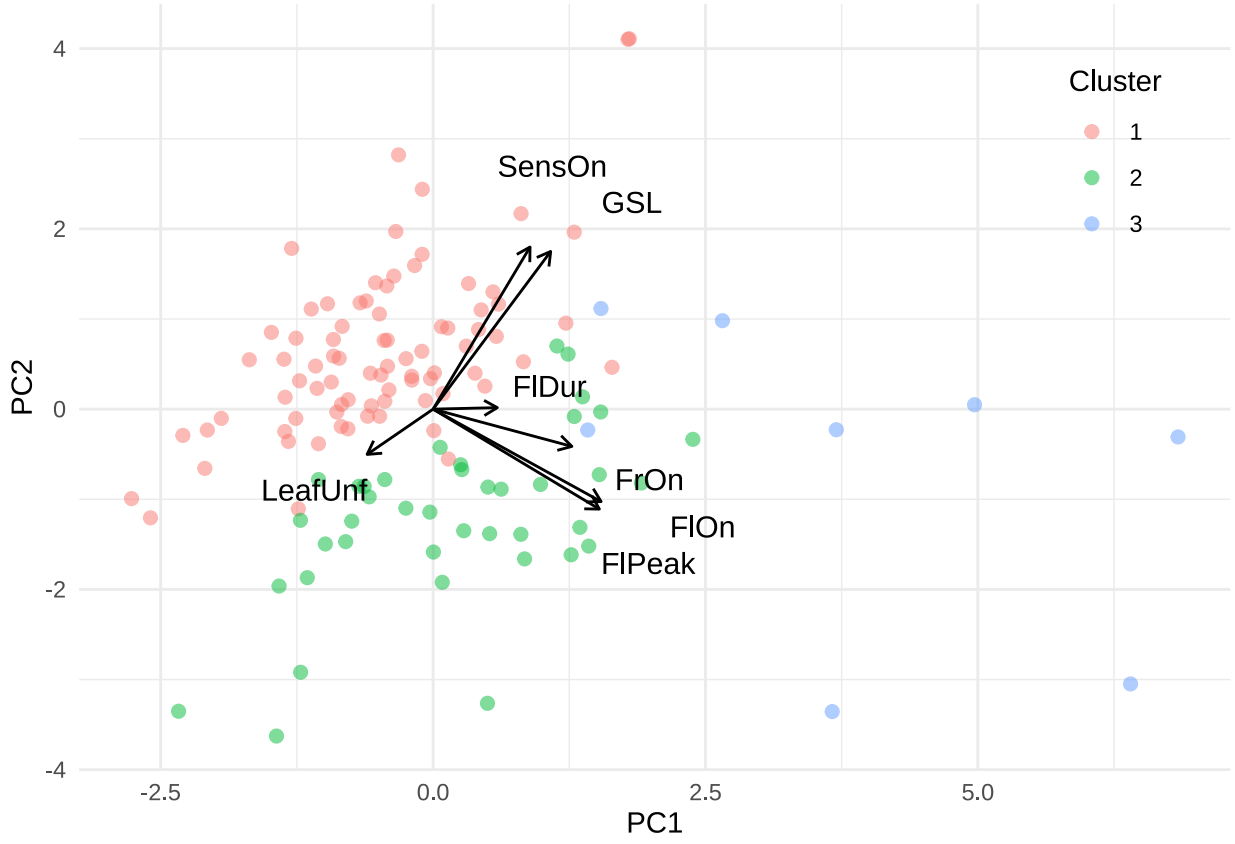

```
scores$cluster <- factor(cutree(hc, k = 3))
ggplot(scores, aes(PC1, PC2, color = cluster)) +
  geom_point(size = 2, alpha = 0.8) +
  stat_ellipse(type = "norm", level = 0.68) +
  theme_minimal()
```



Plot with clusters

```
ggplot(
  scores %>%
    left_join(
      sens_per_endanger %>%
        dplyr::select(species, prop_endangerment),
      by = "species"
    ),
  aes(x = PC1, y = PC2, color = cluster)
) +
  geom_point(alpha = 0.5, size = 2) +
  #geom_text_repel(aes(label = label), size = 3, max.overlaps = 15) +
  geom_segment(
    data = loadings,
    aes(x = 0, y = 0, xend = PC1 * 3, yend = PC2 * 3),
    arrow = arrow(length = unit(0.2, "cm")),
    color = "black",
    inherit.aes = FALSE
  ) +
  geom_text_repel(
    data = loadings,
    aes(x = PC1 * 4, y = PC2 * 4, label = variable),
    color = "black",
```

```
  size = 4,
  inherit.aes = FALSE,
  nudge_x = 0.1* sign(loadings$PC1),
  nudge_y = 0.1 * sign(loadings$PC2),
  box.padding = 0.2,      # distance around text box
  point.padding = 0.2,    # distance from arrows/points
  max.overlaps = Inf
  ) +
theme_minimal() +
labs(
  x = "PC1",
  y = "PC2",
  color = "Cluster"
) +
#stat_ellipse(type = "norm", level = 0.68) +
theme(legend.position = c(0.9, 0.8))
```



RDA effect of endangerment

Endangered or not

```
rda_trait_endanger <- rda(
  sens_per_endanger %>% dplyr::select(starts_with("mean_slope_")) %>% drop_na() ~
    (sens_per_endanger %>%
       mutate(endangerment = factor(
         ifelse(prop_endangerment == 0, 0, 1)))%>% drop_na())$endangerment,
  data = sens_per_endanger
  )
anova(rda_trait_endanger, permutations = 999)
```

```
## Permutation test for rda under reduced model
## Permutation: free
## Number of permutations: 999
##
## Model: rda(formula = sens_per_endanger %>% dplyr::select(starts_with("mean_slope_")) %>% drop_na() ~
##           Df Variance      F Pr(>F)
## Model      1    27.16 4.2421  0.003 **
## Residual 127   813.08
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(rda_trait_endanger, by = "axis", permutations = 999)
```

```
## Permutation test for rda under reduced model
## Forward tests for axes
## Permutation: free
## Number of permutations: 999
##
## Model: rda(formula = sens_per_endanger %>% dplyr::select(starts_with("mean_slope_")) %>% drop_na() ~
##           Df Variance      F Pr(>F)
## RDA1       1    27.16 4.2421  0.004 **
## Residual 127   813.08
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
RsquareAdj(rda_trait_endanger)
```

```
## $r.squared
## [1] 0.03232277
##
## $adj.r.squared
## [1] 0.02470326
```

Proportion of endangerment

```
rda_trait_endanger_prop <- rda(
  sens_per_endanger %>% dplyr::select(starts_with("mean_slope_")) %>% drop_na() ~
    (sens_per_endanger%>% drop_na())$prop_endangerment,
  data = sens_per_endanger
  )
anova(rda_trait_endanger_prop, permutations = 999)
```

```
## Permutation test for rda under reduced model
## Permutation: free
## Number of permutations: 999
##
## Model: rda(formula = sens_per_endanger %>% dplyr::select(starts_with("mean_slope_")) %>% drop_na() ~
##           Df Variance      F Pr(>F)
## Model      1     7.35 1.1203  0.336
## Residual 127   832.89
```

```
anova(rda_trait_endanger_prop, by = "axis", permutations = 999)
```

```
## Permutation test for rda under reduced model
## Forward tests for axes
## Permutation: free
## Number of permutations: 999
##
## Model: rda(formula = sens_per_endanger %>% dplyr::select(starts_with("mean_slope_")) %>% drop_na() ~
##           Df Variance      F Pr(>F)
## RDA1       1     7.35 1.1203  0.327
## Residual 127   832.89
```

```
RsquareAdj(rda_trait_endanger_prop)
```

```
## $r.squared
## [1] 0.008744259
##
## $adj.r.squared
## [1] 0.0009390959
```

Knit document

```
knit_docx(
  template_docx_file = here("docs", "PhenObs_cluster1_draft.docx"),
  knitted_docx_file = here("docs", "PhenObs_cluster1_draft_knitted.docx")
)
```

```
## [1] "Saving knitted document locally"
```

```
doc <- read_docx(
  here("docs", "PhenObs_cluster1_draft_knitted.docx")
)

tables <- list(
  TABLE_1_HERE = tab1,
  TABLE_2_HERE = tab2,
  TABLE_3_HERE = tab3,
  TABLE_4_HERE = tab4,
  TABLE_5_HERE = tab5,
  TABLE_6_HERE = tab6,
```

```
  TABLE_7_HERE = tab7,
  TABLE_8_HERE = tab8,
  TABLE_9_HERE = tab9,
  TABLE_10_HERE = tab10,
  TABLE_11_HERE = tab11,
  TABLE_12_HERE = tab12,
  TABLE_13_HERE = tab13,
  TABLE_14_HERE = tab14,
  TABLE_15_HERE = tab15,
  TABLE_16_HERE = tab16
)

for (ph in names(tables)) {

  doc <- cursor_begin(doc)
  doc <- cursor_reach(doc, ph)
  doc <- body_remove(doc)

  doc <- body_add_flextable(
    doc,
    value = tables[[ph]],
    pos = "on"
  )
}

print(
  doc,
  target = here("docs", "PhenObs_cluster1_draft_knitted_final.docx")
)
```

Session info

```
sessionInfo()
```

```
## R version 4.6.1 (2026-06-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=Spanish_Spain.utf8  LC_CTYPE=Spanish_Spain.utf8
## [3] LC_MONETARY=Spanish_Spain.utf8 LC_NUMERIC=C
## [5] LC_TIME=Spanish_Spain.utf8
##
## time zone: Europe/Madrid
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
```

```
##
## other attached packages:
##  [1] pandoc_0.2.0                  officer_0.7.6
##  [3] flextable_0.10.0              patchwork_1.3.2
##  [5] modelsummary_2.6.0            reproduciblemanuscript_0.1.0.0
##  [7] showtext_0.9-8                showtextdb_3.0
##  [9] sysfonts_0.8.9                units_1.0-1
## [11] rnaturalearth_1.2.0           broom.mixed_0.2.9.7
## [13] broom_1.0.13                  vegan_2.7-5
## [15] permute_0.9-10                corrplot_0.95
## [17] ggrepel_0.9.8                 progress_1.2.3
## [19] countrycode_1.8.0             furrr_0.4.0
## [21] future_1.70.0                 gridExtra_2.3.1
## [23] readxl_1.5.0                  rgbif_3.8.5
## [25] taxize_0.10.1                 ggeffects_2.3.2
## [27] glmmTMB_1.1.14                sf_1.1-1
## [29] lubridate_1.9.5               forcats_1.0.1
## [31] stringr_1.6.0                 dplyr_1.2.1
## [33] purrr_1.2.2                   tidyr_1.3.2
## [35] tibble_3.3.1                  ggplot2_4.0.3
## [37] tidyverse_2.0.0               here_1.0.2
## [39] readr_2.2.0
##
## loaded via a namespace (and not attached):
##   [1] RColorBrewer_1.1-3     rstudioapi_0.19.0      jsonlite_2.0.0
##   [4] datawizard_1.3.1       magrittr_2.0.5         farver_2.1.2
##   [7] nloptr_2.2.1           rmarkdown_2.31         fs_2.1.0
##  [10] ragg_1.5.2             vctrs_0.7.3            minqa_1.2.8
##  [13] effectsize_1.0.3       askpass_1.2.1          htmltools_0.5.9
##  [16] haven_2.5.5            curl_7.1.0             cellranger_1.1.0
##  [19] parallelly_1.48.0      KernSmooth_2.23-26     plyr_1.8.9
##  [22] sandwich_3.1-1         zoo_1.8-15             TMB_1.9.23
##  [25] uuid_1.2-2             whisker_0.4.1          lifecycle_1.0.5
##  [28] pkgconfig_2.0.3        sjlabelled_1.2.0       Matrix_1.7-5
##  [31] R6_2.6.1               fastmap_1.2.0          rbibutils_2.4.1
##  [34] digest_0.6.39          numDeriv_2016.8-1.1    rprojroot_2.1.1
##  [37] textshaping_1.0.5      labeling_0.4.3         timechange_0.4.0
##  [40] httr_1.4.8             mgcv_1.9-4             compiler_4.6.1
##  [43] proxy_0.4-29           bit64_4.8.2            fontquiver_0.2.1
##  [46] withr_3.0.3            S7_0.2.2               backports_1.5.1
##  [49] DBI_1.3.0              performance_0.17.1     MASS_7.3-65
##  [52] openssl_2.4.2          rappdirs_0.3.4         classInt_0.4-11
##  [55] oai_0.4.0              tools_4.6.1            otel_0.2.0
##  [58] ape_5.8-1              zip_3.0.0              future.apply_1.20.2
##  [61] glue_1.8.1             nlme_3.1-169           grid_4.6.1
##  [64] checkmate_2.3.4        cluster_2.1.8.2        generics_0.1.4
##  [67] gtable_0.3.6           tzdb_0.5.0             class_7.3-23
##  [70] data.table_1.18.4      hms_1.1.4              utf8_1.2.6
##  [73] xml2_1.6.0             tables_0.9.35          pillar_1.11.1
##  [76] vroom_1.7.1            splines_4.6.1          lattice_0.22-9
##  [79] bit_4.6.0              tidyselect_1.2.1       fontLiberation_0.1.0
##  [82] knitr_1.51             fontBitstreamVera_0.1.1 reformulas_0.4.4
##  [85] crul_1.6.0             xfun_0.59              stringi_1.8.7
##  [88] lazyeval_0.2.3         yaml_2.3.12            boot_1.3-32
```

```
##  [91] evaluate_1.0.5          codetools_0.2-20        httpcode_0.3.0
##  [94] gdtools_0.5.1           cli_3.6.6               parameters_0.29.2
##  [97] systemfonts_1.3.2       Rdpack_2.6.6            Rcpp_1.1.1-1.1
## [100] globals_0.19.1          parallel_4.6.1          prettyunits_1.2.0
## [103] bayestestR_0.18.1       lme4_2.0-6              listenv_1.0.0
## [106] scales_1.4.0            e1071_1.7-17            insight_1.5.2
## [109] crayon_1.5.3            rlang_1.2.0
```